

Unwatched Pollution: The Effect of Intermittent Monitoring on Air Quality

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Abstract

Compliance monitoring of environmental standards is often conducted intermittently due to its expense. Intermittent monitoring, however, may encourage polluters to show compliance during monitoring, but to increase polluting activities at other times. This paper documents strategic responses to a cyclical, once-every-six-day monitoring schedule under a broad-scale federal air quality regulation. Using an independent satellite-based measure of air pollution, I show that air quality where this cyclical monitoring takes place is significantly worse on unmonitored days. Larger effects are observed in areas that (1) are out-of-compliance with air quality standards, and thus face steeper potential penalties; (2) have a high concentration of certain industries, such as wood product manufacturing, and (3) are intersected by highways. Consistent with this last finding, I show that local governments are more likely to issue air quality advisories that call for citizens to reduce automobile usage and outdoor activities on days when pollution monitoring is scheduled. This evidence points to local governments' coordinating the avoidance of pollution on monitoring days. Finally, I document some consequences of these strategic responses: higher levels of pollution on unmonitored days lead to lower school test scores and elevated levels of violent crime on these days.

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1. Introduction

Enforcement of environmental regulation relies on accurate monitoring of compliance behavior. In practice, limited budgets often force monitoring to be conducted on an intermittent basis. A consequence of intermittent monitoring is that it creates an opportunity for polluters to show compliance during monitoring but to increase polluting activities when no monitoring takes place. The potential for such strategic responses is exacerbated in contexts where the polluters anticipate the regulator's monitoring schedules and have the ability to adjust polluting activities on a short-term basis, as demonstrated by the recent Volkswagen emissions scandal.¹ Strategic responses to intermittent monitoring, however, are generally difficult to detect due to two main challenges. First, independent measurement of polluting behavior during unmonitored times is usually unavailable. Second, monitoring is often endogenous. A simple comparison of monitored and unmonitored polluting activities likely confounds the causal effect of intermittent monitoring with latent factors such as economic conditions and polluters' compliance history.

This paper documents strategic responses to a broad-scale federal air pollution regulation. I explore a unique empirical context where monitoring is based on a publicly available schedule. Under the Clean Air Act, the U.S. Environmental Protection Agency (EPA) is charged with enforcing a national safety level for outdoor particle air pollution (known as particulate matter, henceforth PM) that all counties are required to achieve.² A network of monitoring sites tracks compliance with the standard. Due to high operating and maintenance costs, the EPA grants permission for many sites to monitor pollution on an intermittent basis. To balance between the goal of obtaining representative monitoring results and the administrative costs that would be incurred by a random monitoring scheme, the EPA pre-announces a cyclical, once-every-sixth-day ("1-in-6 day") monitoring schedule each year to be followed by intermittent monitors. I leverage this monitoring policy experiment to examine whether higher concentrations of pollution are observed during days when monitoring is not scheduled. I begin by constructing an indirect measure of particle pollution using 13 years of satellite observations. This measure allows me to observe

¹ See Gates, Ewing, Russell and Watkins (2016).

² Failure to comply with the EPA's outdoor air quality regulation is understood to incur tremendous economic costs for the county. For example, Walker (2013) estimates that the Clean Air Act's 1990 Amendments for particulate matter pollution costs workers about \$8 billion in earnings in the years following non-attainment designations; Greenstone, List, and Syverson (2012) show that ambient pollution regulations led to reduced productivity representing a loss of approximately \$20 billion in revenues annually among U.S. manufacturers. Notably, air quality often improves in non-compliance areas, as substantial resources are spent to limit polluting activities, see e.g., Chay and Greenstone (2003), Auffhammer, Bento, and Lowe (2009).

air quality both during monitor “on-days” and monitor “off-days”. Because the incentive to avoid monitoring is plausibly the only factor that changes on a 1-in-6 day basis, the difference in the level of pollution during off-days and on-days provides evidence of strategic responses to the monitoring schedule.

In the baseline specification, I use the satellite measure to compare pollution levels on off-days and on-days around monitoring sites that follow the 1-in-6 day schedule. The results suggest a significant pollution gap between on-days and off-days. The satellite detects 1.6 percent less particle pollution during on-days than during off-days, while pollution levels between off-days do not differ significantly from each other. Placebo tests show that no such pollution gap is detectable in the absence of an incentive to avoid monitoring. For example, the effect disappears when a monitor retires; and the pollution gap is not found around sites that monitor pollution every day.

Next, I explore sources of the pollution gap to shed light on mechanisms. I begin by documenting substantial cross-sectional variation in the magnitude of the 1-in-6 day pollution gap across counties. Notably, this exercise identifies a number of regional clusters where the pollution gap is exceptionally large (more than 6 percent pollution reduction during on-days). I then present evidence on observable characteristics of these “hot-spot” regions. There are two main sets of findings. First, the correlations are consistent with the intuition on the role of regulatory incentives. For instance, I find that regions out of compliance with the air quality regulations are much more likely to be pollution gap hot spots; this is consistent with the higher regulatory costs borne by non-compliance regions, which give them a stronger incentive to reduce measured pollution levels. Second, hot spots are more common in regions with high concentrations of certain polluting industries, such as wood product manufacturing and mining. While these correlational findings need not reflect causal effects, I provide complementary evidence that the 1-in-6 day pollution gap exists around facilities in relevant industries. Moreover, the pattern is most clearly observed near facilities that are close to 1-in-6 day monitors.

The data also suggest larger pollution gaps are correlated with the presence of highways in the local area. To the extent that traffic responds to the monitoring schedule is unlikely to arise without coordination at a central level, this evidence suggests that local governments might have played a role in coordinating the avoidance of pollution on monitoring days. I test the plausibility of this mechanism in the context of local governments’ strategic issuance of air quality advisories. These advisories call for citizens to reduce outdoor activities and vehicle use to prevent air quality deterioration. I show that these advisories are 10 percent more likely to be issued on days when pollution monitoring is scheduled. While

the results only speak to coordination of public behavior, they raise the possibility that local government coordination might occur in industrial settings as well.

Strategic polluting activities may impose health risks that compromise the intended cost savings from intermittent monitoring. In a companion paper, Zou, Miller, and Molitor (2017) study the health consequences of the 1-in-6 day monitoring schedule. Using Medicare administrative data on the universe of elderly beneficiaries from 2001 - 2011, Zou, Miller, and Molitor (2017) show that higher pollution on the monitor off-day coincides with a significant increase in the elderly mortality rate the following day. Assuming that air pollution would converge to the current on-day level under a counterfactual policy where polluters face monitoring every day, Zou, Miller, and Molitor (2017) conclude that the mortality costs are an order of magnitude larger than the amount of resources saved by the reduced monitoring frequency. Effects of the pollution gap may also extend beyond health consequences. I show two examples in this paper. First, I use test score data from the California High School Exit Examination to show that standardized exam performance is significantly worse when the test date falls on a monitor off-day. Second, using crime records data, I find that criminal activity, especially violent crime, increase significantly on off-days relative to on-days.

This paper has four main contributions. First, I contribute to the growing literature on the economics of environmental regulation monitoring and enforcement, as reviewed by Gray and Shimshack (2011) and Shimshack (2014). In particular, this paper is related to the recent revelation of monitoring evasion in settings such as car exhaust testing (Oliva, 2015; Reyneart and Sallee, 2017) and illegal oil waste dumping (Vollaard, 2017). This paper extends this literature to a setting of a broad scale, national ambient pollution regulation; strong responses are found in various regions across the country, suggesting that strategic responses are much more pervasive than previously documented. Moreover, this paper suggests that polluters' ability to respond to environmental regulations on a short-term basis may be stronger than appreciated by previous literature that focuses on medium- and long-term regional or sectorial substitution of polluting activities (e.g., Becker and Henderson, 2000; Hanna, 2010; Fowlie, Reguant, and Ryan, 2016).

Second, my results highlight the importance of designing monitoring policies in a way that envisions polluters' possible strategic responses. Current evaluation of the desirability of pollution monitoring methods is often from a purely statistical standpoint that does not take strategic responses into account. For example, the idea of using a cyclical monitoring method to obtain representative pollution information date back to at least Akland (1972); subsequent methodological evaluations, many of which prompted

the eventual adoption of the method in actual regulation, typically compare summary statistics (such as mean and dispersion) of a daily pollution sample to the same statistics computed from a *subsample* of days drawn using the cyclical method (e.g., Akland, 1972; Nehls and Akland, 1973; Gilbert, 1987; Rumburg, Alldredge and Claiborn, 2001). My paper shows that the inherent presumption in such evaluations – that air pollution itself does not respond to the choice of monitoring method – can fail in actual implementation. In particular, I show that a cyclical monitoring schedule can lead to significant deviation in pollution levels from the levels observable to the regulator on monitored days.

Third, this paper contributes to a developing understanding of the value of satellite data in regulatory decision making. Partly due to an inherent difference in their missions of operation, cooperation between environmental regulation agencies such as the U.S. EPA and agencies that operate satellite surveillance of atmospheric pollutants, including the National Aeronautics and Space Administration (NASA) and the National Oceanic and Atmospheric Association (NOAA), remains in a nascent stage. While regulators increasingly recognize the value of satellite data in certain fields such as wildfire surveillance (Ruminski et al., 2006; Ichoku, Kahn and Chin, 2012) and air quality forecasting (Kittaka et al., 2004), it is believed that the potential of satellite data is far from being fully exploited (see e.g., Duncan et al., 2014). This paper presents a specific example where the satellites' unique ability to complement intermittency of ground-based pollution monitoring can be leveraged to inform the desirability of a pollution monitoring policy. More broadly, this paper is linked to the growing use of remote sensing data in economic and policy research, reviewed by Donaldson and Storeygard (2016). A closely related study to mine is Grainger (2016), who relies on satellite data to investigate state governments' strategic decisions on the placement of pollution monitors.

Finally, this paper suggests a plausibly exogenous source of air quality variation, and therefore adds to the economic literature on the consequences of air pollution exposure. In particular, my results are related to the developing literature on the effect of short-term air quality fluctuations on cognitive and behavioral outcomes, such as test performance (e.g., Ham, Zweig, and Avol, 2014; Ebenstein, Lavy, and Roth, 2016) and criminal activities (e.g., Herrnsstadt, Heyes, Muehleger, and Saberian, 2016).

The remainder of the paper is organized as follows. Section 2 provides a brief background on particulate matter (PM) regulation and monitoring in the United States. Section 3 presents the main identification of the off-days vs. on-days pollution gap. Sections 4 and 5 explore sources and mechanisms underlying the pollution gap. Section 6 discusses consequences of intermittent monitoring. Section 7 concludes.

2. Background on PM Regulation and Monitoring

2.1. PM Regulation

Regulation of ambient PM (PM_{2.5} and PM₁₀) pollution in the United States is coordinated under the Clean Air Act. The act contains multiple provisions that address the oversight of PM emissions at various industry levels. The regulation on *ambient* PM can be viewed as a policy lever to achieve one of the act's ultimate goals: to maintain outdoor PM concentrations below the established safety levels of the National Ambient Air Quality Standards (NAAQS).³

The Clean Air Act authorizes the EPA to enforce these standards. To make sure that the standards are met nationally, the EPA categorizes counties into “attainment” and “non-attainment” groups based on monitoring results. Non-attainment counties face substantially elevated regulation costs. In cases of non-attainment, the parent state is required to develop a State Implementation Plan that details how plant-specific regulations will be implemented in order to achieve compliance. These regulations typically involve the adoption of pollution abatement technologies and emission limits that affect both existing and new polluters. Local governments and individual polluters receive direct penalties from the EPA in cases of sustained non-attainment. These include financial sanctions that prohibit the approval of almost any highway projects or grants, as well as emission sanctions that require reduced emissions from existing pollution sources for any new or modified emission sources, where the reductions from existing sources must be at least twice the increases from new sources.

Non-attainment regulation poses real regulatory threats. Existing literature has shown that PM non-attainment leads to significant losses in employment and earnings (Walker, 2013), reductions in factories' productivity (Greenstone, List, and Syverson, 2012), and notably, reductions in ambient PM pollution (Auffhammer, Bento, and Lowe, 2009).⁴

2.2. PM Monitoring

³ Separate standards have been established for PM₁₀ and PM_{2.5}, and for four other outdoor pollutants: carbon monoxide (CO), sulfur dioxide (SO₂), nitrogen dioxide (NO₂), and lead.

⁴ A rich literature documents the significant effects of other provisions of the NAAQS targeting at different air pollutants, such as Total Suspended Particulate and Ozone, on air quality (Henderson, 1996; Chay and Greenstone, 2005) and industrial activities (Becker and Henderson, 2000; Greenstone, 2002).

Ambient PM concentration is monitored through a network of more than 1,200 monitoring sites across the country. These sites are usually placed in areas with high population density to ensure a reasonably representative measure of population pollution exposure. During my study period, 2001 to 2013, the PM monitoring network spanned more than 600 counties, accounting for roughly 70 percent of U.S. population.

Unlike monitoring of many gaseous air pollutants (such as ozone) that uses automated laser-based methods, PM monitoring is filter-based and is associated with substantial manual operation and maintenance tasks such as field sample collection and subsequent laboratory analysis. Using the EPA's data on the estimated cost break-down of PM monitoring (U.S. EPA, 1993), Zou, Miller, and Molitor (2017) calculate that the annualized per-site cost of PM monitoring associated with monitor procurement, operation, and maintenance is estimated to be \$21,030 with a 1-in-6 day schedule and \$40,912 with daily (1-in-1 day) monitoring (both estimates in 2013 dollars). With roughly 600 sites operating on a 1-in-6 day schedule, the cost savings from intermittent monitoring aggregates to about \$12 million per year.

High operation and maintenance costs prompt the practice of intermittent monitoring, which has been adopted since the initiation of atmospheric particle pollution sampling in the 1950s (U.S. Public Health Service, 1957). The cyclical 1-in-6 day monitoring method was introduced by Akland (1972) in the wake of regulatory monitoring of total suspended particulates in the 1970s. This practice was subsequently adopted for PM monitoring starting in the 1980s. By that time, the EPA also introduced more frequent 1-in-3 day and daily sampling schedules. These more frequent schedules are used at sites with higher levels of pollution where higher data capture rates are desired (U.S. EPA, 1985). Since the initiation of PM_{2.5} monitoring in the 1990s, the EPA has been tending toward more frequent monitoring. For example, in the initiation of PM_{2.5} monitoring, the EPA requires all PM_{2.5} monitors to operate on a minimum of 1-in-3 day basis (40 CFR 58.13, 1997; U.S. EPA, 1998a). However, in response to states' concerns over cost burdens and a continuing interest in less frequent monitoring, the EPA allows exemptions for lower monitoring frequencies on a case-by-case basis (U.S. EPA, 1998b). From 2001 to 2013, the vast majority of PM monitors followed either a 1-in-6 day (42 percent of monitors), 1-in-3 day (33 percent), or daily (22 percent) schedule.⁵ Although this paper focuses on the 1-in-6 day schedule where gaming is most likely to occur, I also report responses to the 1-in-3 day schedule. In addition, I report "placebo" tests around 1-in-1 day sites where no strategic responses are expected.

⁵ A small number of monitors were granted the exemption to conduct seasonal sampling or, in rare cases, to follow a once-every-12-day schedule.

As a guideline for states to schedule their monitoring routine, the EPA publishes a monitoring calendar on its website at the end of each calendar year, informing states of the monitoring schedule for the next calendar year. Figure 1 presents the calendar for 2001. Notice that the monitoring schedule is not staggered across 1-in-6 day monitors; put differently, all 1-in-6 day monitors are scheduled to sample pollution on Jan 1st, 2001, followed by Jan 7th, 2001 and so forth. Also, starting from year 2001, the 1-in-6 day monitoring schedule follows *strictly* six-day cycles beginning with the first monitoring date of Jan 1st, 2001. Therefore, monitoring status of any day post Jan 1st, 2001 is predetermined. In this sense, the annual calendar publication can be viewed as only providing reminders to states of the monitoring schedule.

3. The 1-in-6 Day Pollution Gap

3.1. Data and Summary Statistics

Monitor Data. I obtain PM monitor characteristics from the EPA's Air Quality System (AQS) for the years 2001 to 2013. The annual summary data files of the AQS are the source of monitor-level information on scheduled number of monitored days, actual monitored days, latitude and longitude location, and annual PM concentration statistics, such as the mean and the max. I identify 1-in-6 day (1-in-3 day, 1-in-1 day) monitors by finding monitors that are required by the EPA to sample 60 or 61 (121 or 122, 365 or 366) days a year.⁶

Satellite Data. I construct a measure for atmospheric particle pollution ("aerosol") using satellite data from the NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) algorithm. Equipped with a flexible set of spectral radiance instruments, MODIS retrieves atmospheric aerosol concentration by measuring the extinction of sunlight, based on knowledge of aerosol's ability to scatter and absorb light at different spectral wavelengths. MODIS summarizes aerosols in a dimensionless index called aerosol optical depth, which has a theoretical range of -0.05 to 5, with smaller value corresponding to lower level of aerosol concentration (Kaufman, et al., 1997; Remer et al., 2005; Voiland, 2010). In the United States, the vast majority of the index's value falls within the range of 0 to 1, with a mean of roughly 0.12. To simplify language, in subsequent sections I refer to this measure as "aerosol" or "aerosol concentration."

⁶ I cross check the validity of this definition by comparing the assigned schedule to the actual daily monitor operation status from AQS' daily monitor summary files, finding that these monitors follow the 1-in-6 day schedule about 95 percent of the time. Note that monitors may occasionally deviate from the monitoring schedule due to power outages and sampler malfunctions. These monitoring compliance status data are reported in the Appendix.

To facilitate interpretation, it is perhaps helpful to proceed with three connections between the concept of the satellite-based aerosol measure and the traditional ground monitor-based PM measure. First, despite the difference in measurement approaches, ground monitor and satellite have a similar *target* of measurement. PM monitors retrieve the concentration of small air particles (e.g. nitrates, sulfates and black carbons) by measuring the amount of particles deposited when air is passed through a size-discriminating filter media; when in the atmosphere, these particles interact with sunlight, and therefore are picked up by the satellite measure. Such an overlap in the measurement target has been the foundation of a large body of atmospheric science literature that documents a strong correlation between the satellite measure and ground-monitoring data.⁷

Second, unlike PM monitors, which measure pollution concentration at the ground level, the satellite orbits at a height of roughly 700 km; when the satellite passes over a given point on the ground, it captures aerosol conditions for the entire column of air from its viewpoint. Therefore, the aerosol measure should be interpreted as the amount of particle pollution in the entire atmosphere corresponding to a given location, rather than a precise measure of population exposure at that location. For example, while the satellite may see a tall smoke stack generating pollution, the population in the local area is probably exposed to that pollution, to a lesser extent, as the wind blows.

Finally, most ground-monitoring data provide the average pollution concentration within a given time period (usually within a 24-hour period) on days when monitoring is conducted; in contrast, satellites provide daily snapshots of pollution in a given area at an (almost) fixed time of day. Aboard the polar-orbiting satellite *Terra*, MODIS continuously observes the earth's atmosphere with a 2,330km-wide sweeping swath, scanning each point on the planet every day at approximately 10:30am local time.⁸ The swaths are then merged together to form daily imagery of aerosol concentration with a spatial resolution of 10km×10km (i.e. half of the size of the average U.S. ZIP Code). My key outcome dataset is a daily panel of aerosol concentration linking each 10km×10km grid cell from 2001 to 2013.⁹

⁷ See e.g., Liu, Franklin, Kahn, and Koutrakis (2007), Lee, Coull, Bell, and Koutrakis (2012), Zhang and Li (2015). Previous economic research has also used the aerosol measure as the proxy for air pollution in developing-country contexts where monitoring data are sparse, see e.g. Foster, Gutierrez, and Kumar (2009), Chen, Jin, Kumar, and Shi (2013), and Bombardini and Li (2016). For a recent review, see Donaldson and Storeygard (2016).

⁸ MODIS is also installed on the satellite *Aqua*, *Terra*'s sister satellite, although data from *Aqua*'s platform is available only since 2004. For that reason, this paper uses data from *Terra*.

⁹ Pixel arrays in the original satellite imagery do not identically overlap on a daily basis. To create the panel dataset, the imagery is mapped onto a fixed map of 10km×10km grid cells, which I obtain from the U.S. National Grid Information Center. This procedure ensures that the grid dataset preserves the original resolution of the satellite imagery, while each grid tracks aerosol concentrations for the same area over time.

Summary Statistics. Table 1 presents satellite aerosol and ground monitor summary statistics by calendar year. Columns 1 to 4 show 10km×10km grid × daily-level satellite aerosol statistics. Over the period of study, the aerosol level stayed relatively stable, declining by an average of 0.5 percent per year from 2001 to 2013.¹⁰

Columns 5 to 7 of Table 1 show monitor-level statistics, including, for each year, the total number of monitors, number of 1-in-6 day monitors, and number of 1-in-6 day monitors exceeding PM NAAQS.¹¹ Columns 5 and 6 show that the total number of monitors decreased over time. This trend began in 1997, with a NAAQS revision that initiated PM_{2.5} monitoring and redirected sources from monitoring of coarser particulates pollution (U.S. EPA, 1997a). As will be described in further detail below, I exploit monitor retirement events to show that the pollution gap between off- and on-days disappears as monitors retire. Column 7 shows that, every year, about 7 percent of 1-in-6 day monitors exceed NAAQS.¹²

Columns 8 and 9 present counts of monitoring site. A monitoring site is a geographic unit that may contain multiple monitors. The main analysis is done at the monitoring site level because the locations of different, individual monitors within the site are not distinguished among one other. To be conservative in aggregating individual monitor-level schedules to the site level, I define a site to be a 1-in-6 day site if any PM monitor in that site follows the 1-in-6 day schedule. Defining the sample this way is expected to bias against finding strategic responses to the six-day monitoring schedule. For example, some monitoring sites may have a daily monitor *and* a 1-in-6 day monitor, where the latter is used to provide quality assurance data for the former.¹³ In the analysis below, I confirm that the evidence of schedule gaming is stronger if I restrict to sites with a standalone 1-in-6 day monitor.

Columns 10 to 13 present monitor-level statistics aggregated to the county level. Column 10 shows the number of counties that have PM monitors. Column 11 shows that a population of roughly 200 million people live in these counties in the lower 48 states, and column 13 shows that about 65 percent of them live in counties that have at least one 1-in-6 day monitor. Note that, although total population coverage

¹⁰ The rate of decline of aerosol is about 1.42 percent per year around PM_{2.5} monitors and about 1.05 percent around PM₁₀ monitors. These are roughly on par with trends of pollution measured using ground monitor data, which show an annual decline rate of 2.3 percent for PM_{2.5} and 1.35 percent for PM₁₀.

¹¹ In all following analyses, I restrict to NAAQS-eligible monitors, i.e., monitors that obtain at least 75 percent of required samples for each quarter of the year. Monitors that fail to satisfy such requirement are not eligible to be used by states to demonstrate NAAQS compliance. Results are not sensitive to the inclusion of ineligible monitors.

¹² The number of NAAQS-exceeding monitors decreases over the sample period, with a temporary increase around 2006 due to a tightening of the PM_{2.5} maximum standards starting October 2006.

¹³ For quality assurance purpose, the EPA requires a certain percentage of 1-in-1 day and 1-in-3 day monitors to be “collocated” with a 1-in-6 day monitor in each state. See more discussion in the Appendix.

of PM-monitoring counties has stayed almost constant over time, population that live in 1-in-6 day counties has declined significantly during the study period, driven by the substantial decrease in the number of 1-in-6 day monitors (column 6).

3.2. Empirical Framework

Empirical Specification. The strictly 1-in-6 day design of the monitoring schedule motivates a straightforward identification strategy that estimates the causal effect of the schedule on pollution by simply comparing levels of air pollution across days of a 1-in-6 day monitoring cycle. The estimation equation is

$$Aerosol_{st} = \sum_{d=-3, d \neq 0}^2 \beta_d \cdot 1(t = d) + Time_t + \alpha_s + X_{st}\gamma + \varepsilon_{st} \quad (1)$$

where $Aerosol_{st}$ is the logged satellite aerosol concentration at monitoring site s at time t , measured by the daily aerosol level within the 10km×10km grid cell that corresponds to the land area containing the site.¹⁴ The key coefficients of interest are the β 's ($\beta_{-3}, \beta_{-2}, \beta_{-1}, \beta_1, \beta_2$) that represent air pollution on each day of cycle, running from three days before to two days after the on-day. The on-day is marked as day 0, which is the omitted category in the regression, so the β 's should be interpreted as percentage changes in air pollution during the off-days relative to the on-day. The strictly 1-in-6 day cyclicity of the off-days treatment implies that very few confounders may bias β 's from identifying the causal effect of the monitoring schedule, especially given the extensive length of the panel dataset. To confirm this point, I report results from two types of specifications. In the first, I report estimates of β 's conditional on no covariates, so that β_d simply shows the raw difference between pollution on day d of a cycle relative to the on-day. Second, I report regressions that include a rich array of controls including time fixed effects $Time_t$ (year, month-of-year, and day-of-week fixed effects), monitoring site fixed effects α_s , as well as X_{st} , which is a matrix of time-variant weather controls including daily temperature categorized into ten 10-degree bins, daily wind speed quartiles, and quadratic daily precipitation. Since pollution observed at a

¹⁴ For the sake of description, in the main analysis I ignore the fact that a 10km×10km grid may contain multiple monitoring sites. In fact, during the study period the most populous grid contains 13 monitoring sites. However, more than 85 percent of grids contain a single site, and less than 1 percent of grids contain more than three sites. I confirm that dropping duplicative grid-day observations has negligible impacts on the results.

site is likely driven by emissions elsewhere that also affect nearby sites, all inferences allow for correlations in errors across different monitoring sites within the same county, clustering standard errors at the county level.

I also estimate a more parsimonious version of equation (1) which takes the following form

$$Aerosol_{st} = \beta \cdot 1(Off-days_t) + Time_t + \alpha_s + X_{st}\gamma + \varepsilon_{st} \quad (2)$$

All components in this estimation equation are the same with equation (1), with the only difference being that, rather than having five dummies separately indicating days of a 1-in-6 day monitoring cycle, equation (2) includes the $1(Off-days_t)$ dummy which indicates all five off-days. The coefficient β therefore represents the gap in pollution levels between an average off-day vs. an average on-day.

To interpret β as the causal impact of the monitoring schedule on air pollution, the identification assumption must hold that no differential pollution levels would have been observed between on- and off-days in the absence of the monitoring schedule. Put differently, I assume that the only reason that ambient air quality might show a significant pattern once every six days is because polluters react to the incentive of monitoring avoidance generated by the 1-in-6 day sampling schedule. While this assumption is not directly testable, I report two types of “placebo” tests to support the plausibility of this assumption. First, I exploit about 490 cases of monitoring site retirement from 2001-2013, contrasting 1-in-6 day pollution gaps before and after site retirement. The underlying idea is that, if gaming is targeting at the sampling schedule, then no pollution gap should be observed in the same place following the retirement of a 1-in-6 day monitoring schedule. Second, I repeat the 1-in-6 day pollution examination in areas with “placebo” sites, where gaming is not expected. These include sites that follow the 1-in-1 day schedule, as well as sites that do follow the 1-in-6 day schedule, but for the monitoring of air toxicants not directly linked to any regulation standards.

Interpretation: changes versus levels. The estimation framework above identifies the causal effect of intermittent monitoring on strategic responses by looking at *changes* in pollution levels on off-days relative to on-days (“Is off-days pollution higher than on-days?”). The framework, however, does not identify the effect of intermittent monitoring on the average *level* of pollution. For example, the research design does not speak to the counterfactual levels of air pollution if places that currently undertake 1-in-

6 day monitoring conduct 1-in-1 day monitoring instead (“If polluters face 1-in-1 day monitoring instead, will pollution levels on off-days reduce to the current on-day level?”). While this paper primarily focuses on studying strategic responses, knowledge of counterfactual pollution levels becomes important when thinking about the social damages, such as health costs, due to intermittent monitoring. I discuss this issue further in section 6.

3.3. Baseline Results: 1-in-6 day Pollution Gap

I first estimate equation (1) using my preferred sample, which includes all monitoring sites containing at least one 1-in-6 day PM monitor from 2001-2013. The sample includes 1,193 monitoring sites that span 563 counties in the lower 48 states. Figure 2 reports the results. I do not condition the regression on any covariates, so the solid line simply represents the time path of air pollution in a 1-in-6 day monitoring cycle, averaged across all cycles in the sample. As described in the introduction, within a typical monitoring cycle, air pollution exhibits a flat path, except for a sharp drop during the on-day. This pattern provides a striking revelation of polluters’ ability to manipulate ambient air quality at the monitoring sites on a short-term basis.

Table 2 reports the average 1-in-6 day off-days vs. on-days pollution gap using equation (2). Results in column 1 correspond to Figure 2 and show that air pollution is on average 1.6 percent higher on an off-day relative to an on-day. Column 2 reports that adding the full set of controls does not change the estimates. In column 3, I restrict the estimation sample to sites with a standalone 1-in-6 day PM monitor. This action is expected to reduce the diluting impact from sites where the 1-in-6 day monitor is collocated with high-frequency monitors, such as ones that sample every day. In this case, the pollution gap rises to about 1.8 percent. The effect persists if I further restrict the sample to counties with only 1-in-6 day monitors (column 4). The fact that gaming on average appears stronger in standalone sites provides suggestive confirmation that gaming is indeed targeting 1-in-6 day monitors.¹⁵

3.4. Placebo Tests

¹⁵ Coefficient estimates obtained using the preferred sample and the restrictive sample are not statistically distinguishable. A joint test of equal effects across sites with multiple monitors and a standalone 1-in-6 day monitor yields a p -value of 0.448.

The identification assumption states that no pollution gap would have been observed in the absence of the 1-in-6 day monitoring schedule. To boost the confidence in the internal validity of the empirical design, I provide two types of placebo tests that establish a null 1-in-6 day pollution gap in places where gaming is not expected.

The first test explores the retirement of 1-in-6 day monitoring sites. If gaming is indeed targeting the 1-in-6 day schedule, then one should expect the disappearance of the pollution gap after sites are removed.¹⁶ To operationalize this test, I first draw upon information in the EPA's monitor listing file and identify 490 cases of 1-in-6 day monitoring sites retirement events. The analysis then uses the satellite measure to track air quality in the areas where these sites are located, and compares the off-days vs. on-days pollution gap before and after sites' retirement. Note that I can estimate the pollution gap even after the site was removed because the monitoring calendar is universally applied, and, hence, even in the absence of a working monitoring site, I know what the sampling dates would have been. Figure 3 reports the results, where the 1-in-6 day pollution gap is shown as a function of years relative to sites' retirement. The gap is about 2.1 percent for the time frame when the site was still operating; for the exact same area, the gap closes immediately after monitor retires. This pattern provides support for the hypothesis that gaming is targeting at the monitoring schedule.

In the second type of placebo check, I apply the same methodology to estimate the 1-in-6 day pollution gap near sites where gaming is either not feasible or not necessary. These include about 560 monitoring sites that follow the 1-in-1 day schedule, and about 800 hazardous air pollutants (HAPs) monitoring sites that also follow a 1-in-6 day schedule for the monitoring of other pollutants not subject to any regulatory standards.¹⁷ Table 3 reports that no significant 1-in-6 day pollution pattern is detected near these sites. Table 3 also reports a simple power calculation that shows that the placebo tests have enough statistical power to detect an effect that is similar in size to the effect found in the main analysis (Table 2) at a conventional significance level. These findings again support the identification assumption that no 1-in-6 day pollution gap would have been observed in the absence of the 1-in-6 day monitoring schedule.

¹⁶ Monitoring sites retire for two main reasons: (1) budgets are reallocated to the monitoring of other pollutants; (2) pollution sources in the monitored area exit, leaving little need to continue monitoring. These reasons are not distinguished in the test which simply exploits the discrete removal of the incentive to respond to 1-in-6 day monitoring.

¹⁷ These sites monitor a total of 734 different toxic air pollutants among which the five most commonly monitored are Benzene, Toluene, Ethylbenzene, o-Xylene, and Styrene.

3.5. Pollution Gap Near 1-in-3 day Monitoring Sites

Are strategic responses observed around 1-in-3 day monitoring sites? I now repeat the same analysis for sites that follow the 1-in-3 day monitoring schedule. Despite the relatively high sampling frequency, it is an empirical question whether polluters can engage in effective gaming against a sampling schedule operating on a one-in-three-day basis.

My analysis finds no strong evidence of gaming against the 1-in-3 day sites. Figure 4 shows that, on average, the pollution path within a typical 3-day monitoring cycle exhibits a “V” shape, but the difference in pollution levels on off-days vs. on-days is not significant. Regression estimates in Table 4 confirm this finding. Columns 1 and 2 show that the average 1-in-3 day pollution gap is about 0.3 percent and is not statistically significant whether or not control variables are included. Column 3 reports that restricting to sites with standalone 1-in-3 day monitors does not increase the estimate. When the estimation sample is further restricted to counties with only 1-in-3 day monitors (column 4), I find a pollution gap of 0.54, and it is marginally significant ($p = 0.076$). Again, a simple power calculation suggests sufficient statistical power to detect an effect size similar to the 1-in-6 day pollution gap (power > 0.999 for detection of a 1.5 percent effect that is significant at the 5 percent level).

4. Sources of the Pollution Gap

This section explores correlates of the pollution gap to shed light on the sources of the 1-in-6 day pollution gap. Section 4.1 begins by documenting the cross-sectional variation in the magnitude of the pollution gap across counties. In section 4.2, I present a data-driven exercise to explore economic and industrial characteristics of regions with large pollution gaps.

4.1. Identifying Pollution Gap “Hot Spots”

Pollution Gaps across Counties. I begin by estimating one off-day vs. on-day pollution gap per county in the contiguous U.S. For each county c , the following estimation equation is fitted:

$$Aerosol_{i(c)t} = \beta_c \cdot 1(Off-days_t) + Time_t + \alpha_{(c)} + \varepsilon_{i(c)t} \quad (3)$$

where $Aerosol_{i(c)t}$ denotes logged aerosol level in grid i inside county c on date t . Seasonality controls $Time_t$ include year, month-of-year, and day-of-week dummies. County fixed effects $\alpha_{(c)}$ are not actually included in the estimation, because regressions are run separately by county. The average county-level regression contains 35,236 observations (median = 21,086 observations) at the 10km×10km-by-daily level.¹⁸

Figure 5 plots $\hat{\beta}_c$'s, the county-level estimates of the pollution gap from equation (3). The map is drawn so that warmer colors indicate areas where pollution is higher during monitor off-days relative to on-days. For reasons that I will explain below, more visual emphasis is given to counties near pollution gap “hot spots,” defined as counties in the top decile in terms of the magnitude of the pollution gap.

The map exhibits two features. First, areas with large pollution gaps exhibit a strong “clustering” pattern that is evident in parts of California, parts of Montana, Southern Texas, as well as a group of states in the Midwestern U.S. Partly due to this spatial nature, most conclusions I present in this section are not qualitatively sensitive to alternative definitions of hot spots, such as defining hot spots as counties with pollution gaps in the top quintile. Second, some counties exhibit a *negative* pollution gap, which indicates that aerosol levels are *lower* during off-days in comparison to on-days. These areas are usually observed in between hot spots, and are more common in the eastern United States. In the following discussion, I show that wind transport can provide a potential explanation for negative pollution gaps.

Wind Transport and Pollution Gap Shifts. Wind transport of air pollutants can create complex dynamics, and therefore complicates the interpretation of the magnitude of 1-in-6 day pollution gaps across counties. Two complications may arise. First, even if a county does not respond strategically to the monitoring schedule, it may exhibit a pollution gap if it locates near a county that does respond. Second, because the pollution gap is a measure of pollution differences *relative* to the on-days pollution levels, pollution transport may cause a positive pollution gap in the “source” county to appear as a negative gap in other counties.

¹⁸ Alternatively, one could employ a county fixed effects approach and obtain the same point estimates of β_c in a single regression where the off-days dummy and the time fixed effects are allowed to vary flexibly by county; such a procedure, however, will substantially increase the computational burden. Results are very similar in estimating equation (3) with alternative levels of stringency, such as removing all controls or further including for grid cell-specific effects in addition to seasonality controls.

Figure 6 presents a test of whether the geographic pattern of pollution gap (Figure 5) is consistent with such dynamics. In this heat chart, a color cell represents the pollution gap estimate for a given group of counties, with warmer color indicating larger pollution gap. Row-wise, the heat chart shows pollution gap estimates by counties in each decile of distance to hot-spot counties, starting from the hot-spot counties themselves on the bottom row. Column-wise, the chart shows estimates from the exact same data, but using different days of cycle for normalization. Take color cells on row “94” for an example. This row shows pollution gap estimates for the decile of counties that are on average 94 miles away from hot-spot counties; the color cell in column 0 (column 1) shows the pollution gap estimate when pollution on day 0, i.e., the on-day (day 1, i.e., the day following the on-day) is normalized to zero.

The graph exhibits three patterns: (1) begin with column 0, where pollution gap is defined by normalizing the on-day to zero. This shows that pollution gaps in hot-spot counties are large, but gradually turns smaller, and eventually become negative, as one moves away from the hot spots; (2) these negative pollution gaps in column 0 appear as positive gaps in some other columns where the normalization is based on alternative days of cycle. Notably, the solid arrows show that the timing of “pollution-decline-day”, indicated by the warmest color cell within each row, shifts systematically to the right. As a result, the visual pattern shows a “plume” like shape; and (3) the speed of the pollution gap shifts is consistent with the average vector wind speed in the United States of 3 miles per hour. For example, the dashed arrows show the predicted path of pollution gap shifts, where a 1-in-6 day pollution gap is assumed to be generated by the hot-spot counties and shifts at a speed of 3 miles per hour.

In the Appendix, I report additional analysis that further disaggregates the wind-transport analysis by taking into account counties’ upwind/downwind status relative to the hot spots. I find a strong shifting pattern of the pollution gap in downwind counties, and little shifting in upwind counties. In subsequent sections, I focus primarily on characteristics of hot spots and set aside the variation in the magnitude of the pollution gap, because the latter is more likely to reflect a complex mix of local strategic responses and a blown-in pollution pattern from neighboring areas.

4.2. Characteristics of the Pollution Gap Hot Spots

My previous analysis uncovered substantial variation in the 1-in-6 day pollution gap across counties. Here, I explore cross-sectional correlates of the pollution gap to shed light on potential mechanisms. The empirical exercise seeks to determine which economic and industrial characteristics are explanatory of

counties being pollution gap “hot spots,” defined as those with pollution gap estimates in the top decile of magnitude.

Compliance Status and Emission Concentration Characteristics. While the analysis is ultimately a data-driven exploration of the roles of a broad range of characteristics, I begin with a set of characteristics whose expected direction of correlation is informed by the incentive structure of regulation. These characteristics are:

(1) A county’s $PM_{2.5}$ or PM_{10} NAAQS non-attainment history. As discussed in section 2.1, non-attainment counties bear higher regulation costs and have a higher incentive to reduce measured air pollution levels. Strategic responses therefore are expected to be more likely in these counties. I assign each county its non-attainment indicator based on whether any $PM_{2.5}$ and PM_{10} non-attainment status was designated by the EPA from 2001 to 2013.¹⁹

(2) Presence of 1-in-6 day PM monitoring sites. To the extent that gaming is targeting the 1-in-6 day monitoring schedule, counties with 1-in-6 day PM monitors are expected to have a higher likelihood of being a pollution gap hot spot. Because the monitoring network changes over time, 1-in-6 day monitoring counties are defined as those ever had any standalone 1-in-6 day PM monitors from 2001 to 2013;

(3) County’s emission share concentration (Herfindahl Index). In the context of ambient pollution manipulation, it is reasonable to think that strategic responses should be larger in areas with major polluters than in areas where total emissions are scattered among many smaller polluters. I construct a county-level air pollutants emission Herfindahl Index using annual plant-level emission observations drawn from the EPA’s Toxic Release Inventory (2001-2013). This measure ranges from 0 to 1, where higher value represents counties with higher emissions concentrations. In the subsequent analysis, a county is said to have high emission concentrations if the Herfindahl Index is greater than 0.9.²⁰

¹⁹ While non-attainment is an intuitive measure of a county’s *overall* incentive to respond, in the appendix I report two sets of additional analyses that may better capture *marginal* incentives: (a) in county level cross-section estimation, counties in the top quintile of average $PM_{2.5}$ level is about 5 percentage point more likely to be a hot spot than counties in other quintiles; (b) in monitoring site level panel estimation, the magnitude of the pollution gap is positively correlated with a county’s measured $PM_{2.5}$ level in the previous month, especially when the $PM_{2.5}$ level is close to the non-attainment threshold.

²⁰ Results are robust to using alternative cuts of the Herfindahl Index, such as defining high concentration as having a single polluter in the county, or using the Index directly as the explanatory variable. I also could have built the Herfindahl Index using plant-level PM emission measures from the EPA’s National Emission Inventory (NEI) available every three years since 2008. These additional results are available upon request.

Table 5 presents the association between a county's hot-spot status and the characteristics described above. Columns 1 to 3 report bivariate specifications where the hot-spot indicator is regressed on the characteristics one by one. Columns 1 and 2 show that both non-attainment and 1-in-6 day monitor presence are positively and statistically significantly correlated with hot spots. Both correlations are large in magnitude. Non-attainment status is associated with a 5.2 percentage point increase in the chance of being a hot-spot county, while having 1-in-6 day monitors is associated with a 3.9 percentage point increase in the likelihood. Both correlation estimates are based on a dependent variable mean of 10 percent, reflecting the definition of hot spots in terms of the decile with the largest pollution-gap estimates. On the other hand, column 3 shows that the correlation between hot spots and the Herfindahl Index is insignificant and small in magnitude. Because counties' regulatory and emission characteristics are likely to correlate with each other, I also report a joint regression where the three characteristics are used to jointly explain hot spots. Column 4 suggests that the correlational estimates are similar to those from columns 1 to 3.

While results in columns 3 and 4 suggest a weak correlation of the emission Herfindahl Index with hot spots, I cannot rule out the possibility that the role of emission concentrations may differ across areas. For example, heterogeneity may arise if emission concentrations only matter in areas with a higher propensity to respond to the monitoring schedule. Consistent with that view, column 5 of Table 5 reports that the Herfindahl Index is a very strong predictor of hot spots in counties with 1-in-6 day monitors - as shown by the coefficient on the interaction term between a high Herfindahl Index and the presence of a 1-in-6 day monitoring schedule. In the Appendix, I report additional evidence that a larger pollution gap is indeed observed around 1-in-6 day monitoring sites located in high Herfindahl Index counties. As an important note to interpretation, the results on the role of emission concentrations is consistent with the view that ambient air quality is easier to manipulate in areas with major polluters. However, the results do not necessarily imply that strategic responses are initiated and coordinated among major polluters themselves. As I discuss in further detail toward the end of this section, these findings can alternatively be explained by the fact that it is easier for the local governments to coordinate monitoring avoidance in areas with high concentrations of emissions.

Industry Characteristics. Because industries differ across various dimensions such as PM emission intensity, location, production technology and so forth, they may respond differently to the intermittent monitoring schedule. Here I explore industry determinants by linking pollution gap hot spots to variation in industry concentration. The conceptual exercise is to compare the maps of pollution gap hot spots and

industry concentrations. A stronger industry correlation is therefore indicated by a better line-up between hot spots and counties with a high concentration of that particular industry. I implement this exercise in a simple statistical learning framework. Begin with the following estimation equation:

$$1(Hotspot_c) = \sum_j \alpha_j \cdot \underbrace{1(Industry_c^j)}_{\substack{\text{High concentration} \\ \text{of industry } j \\ \text{in county } c}} + \underbrace{\widehat{X}_c}_{\text{other controls}} \cdot \gamma + \varepsilon_c \quad (4)$$

The key explanatory variables are a set of dummies $1(Industry_c^j)$, each indicating whether county c is in the top decile of industry j 's county-level concentration distribution, where concentration is measured by the share of industry- j employment relative to the county total. In other words, for each industry j , I compute its employment share in each county c , and $1(Industry_c^j)$ indicates counties in the top decile for that industry. To construct the employment share measures, I use 3-digit NAICS-level employment data from the Census County Business Pattern. All measures are computed as averages from 2001 to 2013. The key coefficients of interest are α_j 's, which tell us if a high concentration of industry j is predictive of hot spots, conditional on other industry profiles in the county. Other county-level characteristics X_c include the 1-in-6 day monitoring dummy, the PM non-attainment status, and other controls, such as state fixed effects dummies, described in more detail below.

While the exercise does not aim to tease out the causal relationship between hot spots and industry concentrations, I explore robustness of industry correlates through a variety of specification changes. One concern with the model specification is omitted characteristics. In my primary specification of equation (4), I include all “polluting” industries, defined as 3-digit NAICS industries that belong to sectors (i.e., 2-digit NAICS industries) which contribute to at least 1 percent of national total PM₁₀ emissions.²¹ While this includes more than 30 3-digit industries, one might worry about omitted industries as well as other unobserved correlates that matter for hot spots. I address this concern in a number of ways. First, I present an augmented version of equation (4) where, in addition to polluting industries, I control for dummies indicating high concentrations of all available 3-digit industries included in the County Business Pattern.

²¹ Emission shares are computed using the EPA's 2011 National Emissions Inventory which contains a near-census of all pollution sources known to the EPA from its various regulatory programs.

To capture potential influence from mobile pollution sources, I also include dummies for whether highways and/or major (Class 1) railroads intersect the county. Second, I report models that use within-state variations in industrial concentrations by including state fixed effects. This action is expected to reduce the influence of unobserved geographic correlates that don't vary much within a confined area. Third, I estimate the model in subsamples that are restricted to counties that fall within 50 miles of the hot spots. In a related specification, I further restrict to hot-spot counties themselves, and use the same estimation model to predict the intensity of the pollution gap. This estimation uses $\hat{\beta}_c$ from equation (3) as the outcome variable while weighting observations by the precision measure $1/SE_c^2$ where SE_c is the standard error for the pollution gap estimate $\hat{\beta}_c$.

Another concern with the modelling strategy is overfitting. While the dummy approach enjoys flexibility, having too many explanatory variables can potentially make the estimation inefficient and reduce the generalizability of the results. To alleviate this concern, I pair OLS estimation with the Least Absolute Shrinkage and Selection Operator (LASSO) which performs variable selection on the entire set of regulatory, industrial, and state dummies (Tibshirani, 1996). I employ a simple version of LASSO which regularizes the complexity of the model by imposing a penalty on the sum of absolute values of all regression coefficients. This procedure yields a sparse solution of the original optimization problem by shrinking coefficients of certain explanatory variables to zero.²²

Figure 7 offers a visualization of the results. In this heat chart, a cell represents t -statistics of a coefficient estimate; cells on a same column are obtained from a joint regression which is specified by the chart's header section. The color scheme is such that redness indicates positive correlation; blueness indicates negative correlation; and darker color indicates stronger correlation. I organize polluting industries into sector blocks, ranked from sectors that emitted the highest share of PM (Utilities) to the lowest (Administrative, Support, Waste Management, and Remediation Services). In the interest of space, I do not report coefficients for all other non-polluting industries that contribute less than 1 percent of total PM emissions.

Begin with the panel on the left-hand side, which corresponds to OLS estimation. There are two evident color patterns. First, strong signals (i.e., dark red) are observed from the upper part of the chart, while signals from the lower part of the chart are weak. This pattern is consistent with the view that

²² Conceptually, this imposes a constraint of $\sum_p |\gamma_p| \leq \lambda$ when solving the original OLS problem for equation (4). This would yield a path of solution depending on the degree of penalization λ , and I choose the optimal λ using a 10-fold cross validation.

strategic responses are expected to come from polluters that emit high volumes of particulate matter pollution and therefore have a chance to manipulate ambient air quality. Second, a number of polluting industries exhibit consistently strong positive correlations with hot spots. These include wood product manufacturing, chemical product manufacturing, and mining. Moreover, counties with highway segments are more likely to be pollution gap hot spots (I discuss this finding in further detail in the next section). The LASSO estimation results on the right hand side panel largely confirm the OLS findings. Industries with low emission contributions tend not to be selected as relevant predictors; on the other hand, coefficient estimates remain largely unchanged for industries that are recognized by the OLS as consistent and significant correlates.

Complementary Evidence at the Facility Level. The results thus far point to the number of polluting industries, such as wood product manufacturing, as a strong correlate with pollution gap hot spots. In light of this finding, I use the satellite pollution measure to test if a 1-in-6 day pollution gap is indeed observed around plants in these industries. To further attribute this behavior to monitor-schedule gaming, I estimate a flexible specification that allows the pollution gap to depend on the plant's distance to the nearest 1-in-6 day monitoring site. Because polluters are not expected to have control over ambient air pollution over long distances, gaming is expected to decrease over the distance between the polluter and the monitoring site. The specific estimation equation is as follows

$$Aerosol_{pt} = [1(Off-days_t) \times 1(Dist_{pt} = d)]' \cdot \beta_d + Time_t + \alpha_p + X_{pt}\gamma + \varepsilon_t \quad (5)$$

The outcome variable $Aerosol_{pt}$ is the log aerosol level of the 10km×10km area that contains plant p on date t . The indicator function $1(Dist_{pt} = d)$ is a categorical variable that indicates distance bins from plant p to the nearest 1-in-6 day monitoring site.²³ Interacting this categorical variable with the off-days dummy $1(Off-days_t)$ allows the key coefficient β to vary by distance bin d . In implementation, I focus on all plants that are within 50 miles of monitoring sites, and I group them by distance deciles. All other plants are pooled into a single "> 50 miles" group. Therefore, the β_d will have 11 entries, tracing out a

²³ Note that the distance measure has a time subscript as a plant's distance to the nearest monitor can change if monitors retire.

“distance gradient” of the -in-6 day pollution gap. The rest of the notation is analogous to that in equation (1), except that X_{pt} is understood to include the main distance bin variables $1(Dist_{pt} = d)$.²⁴

In the interest of space, the analysis focuses on the wood product manufacturing industry, which emerges as the strongest industry that correlates with hot spots in the aggregate-level analysis. In the Appendix I report examinations for other sectors, including chemical product manufacturing and mining, and for the presence of highways. The results for these sectors are qualitatively similar to the wood industry.

I draw location information of wood product manufacturing plants from the EPA’s annual Toxic Release Inventory (TRI) data. The 1986 Emergency Planning and Community Right-to-Know Act (EPCRA) requires facilities in selected industries, including wood product manufacturing, to report to the TRI if they have at least 10 full-time employees, and if they process or use any one of EPCRA-listed toxic pollutants by more than a threshold amount. The final estimation sample contains 988 plants in the wood product manufacturing industry.

Figure 8, panel A plots the β_d estimates from equation (5). Results show clear evidence that for plants that are the closest to 1-in-6 day monitoring sites (i.e., plants that fall in the first decile of distance distribution with the average distance equal to 1.9 miles), air quality on an off-day is roughly 4 percent worse than on an on-day, according to the satellite measure. By contrast, this gap is not observed for plants further away from monitoring sites. While the distance gradient curve exhibits average off-days vs. on-days pollution difference, Figure 8, panel B reports the 1-in-6 day pollution pattern for plants in the first decile as well as for plants in the other distance deciles. Around plants close to monitoring sites, the pollution path appears to be flat except for a ramp-up on the day preceding the scheduled monitoring day, followed by a significant decrease during the monitored day itself. This pattern sheds some light on the operative nature of gaming. It suggests that, at least among wood product manufacturers, the gap appears to be explained by substitution between emissions on the day before and during monitoring.

Interpretation. While the empirical exercise provides a clear characterization of hot-spot counties’ industry correlates, it provides minimal answers regarding *why* some industries may respond strategically to the monitoring schedule while others may not. A capacity-related explanation seems reasonable for certain industries. For example, while the utility sector is a major PM emitter, power plants, especially

²⁴ In practice, a 10km×10km grid area may include multiple plants. To avoid repetitive observations, the data are usually collapsed to the grid × daily level before the estimation is done. I have confirmed that results are not sensitive to whether estimation is run at the plant or at the grid level.

coal plants, often run around the clock; for them, ramping up and down production in response to short-term monitoring schedule is very costly. By contrast, wood product manufacturers also contribute to a significant 3 percent of total PM emissions among point sources, but usually run on a low-capacity factor. In fact, wood plants, which operate at roughly 60 percent of capacity, have the lowest utilization rate among all polluting manufacturers examined in this study. The unused capacity might enable these plants to shift production activities around in avoidance of the pollution monitoring schedule.²⁵

Results thus far also do not directly speak to the organization of strategic responses. There are at least two mechanisms through which strategic responses may be operating. Polluting facilities that face the threat of non-attainment regulation may self-initiate strategic responses. Alternatively, one can imagine a setting where local governments coordinate the avoidance of federal pollution monitoring by encouraging reductions of pollution during monitored days. The data are not equipped to distinguish between these two mechanisms. For instance, the fact that larger responses are correlated with a Herfindahl Index showing high emission concentrations in certain counties does not necessarily imply that the effect works through major polluters' self-initiated responses. Instead, it may simply reflect the fact that it is easier for the local governments to coordinate monitor avoidance in such areas. Similar arguments can be used to explain industry findings. Nevertheless, some evidence points to a potential role of government coordination. Most notably, Figure 7 suggests that pollution gap hot spots consistently correlate with the presence of highway segments in the county. To the extent that traffic responses to monitoring schedules are unlikely to arise without manipulation at a central level, the evidence is suggestive of a role of coordination. I provide further evidence on this topic in the following section.

5. An Example of Local Government Coordination: Strategic Air Quality Advisory Issuance

Previous analysis hypothesizes a role of local government coordination in strategic responses to the EPA's PM monitoring schedule. To test such possibility, I focus on an empirical setting where a specific form of coordination behavior is observable. Throughout the country, many local air pollution control agencies adopt air pollution "Action Day" programs that issue public warnings when ambient pollution

²⁵ Capacity utilization statistics are drawn from the Census Bureau's 2008-2013 Quarterly Survey of Plant Capacity Utilization. The wood industry's low capacity utilization rate is partially explained by the fact that it is usually a hard hit during recessions. Capacity utilization for wood (entire manufacturing) dropped from 70 percent (78 percent) at the beginning of 2008 to a lowest 48 percent (67 percent) in the second quarter of 2009.

concentration is expected to reach unhealthy levels.²⁶ When an action day is announced, citizens are advised to “take actions,” such as reducing energy and automobile use, to prevent deterioration of air quality. Previous research shows that these advisories affect outdoor activities and transportation decisions (Neidell, 2009; Cutter and Neidell, 2009; Graff Zivin and Neidell, 2009). I obtain all Action Day records from the EPA’s AirNow program. From 2004 to 2013, 346 reporting areas voluntarily adopted Action Day programs. These areas are a mix of cities, counties, metro areas, and states covering roughly 50 percent of the U.S. population. To avoid double counting issuances in cases of overlapping or nested jurisdictions, I aggregate the data to the core-based metro area (CBSA) level.²⁷ This gives me a total of 14,945 issuances at the CBSA × daily level from 2004 to 2013.

I examine whether Action Day advisories are more likely to be issued on days when PM monitoring is scheduled. The hypothesis is tested using the same estimation framework outlined in equation (1), with the outcome variable changed to a dummy that indicates whether an Action Day advisory is issued on the CBSA × day. Since Action Days are often issued on an “episode” basis spanning multiple days, to better capture issuance timing I also estimate a variant of the model where only the first issuance (6,232 out of 14,945 total Action Days) is count for a period of consecutive Action Days.

The main result is summarized in Figure 9, panel A, which tests for strategic timing of Action Day declarations adjusted for consecutive issuance. The graphical pattern provides evidence of a significant excess of advisories that are issued on pollution sampling days. On average, an action day is 0.10 percentage points more likely to be issued on an on-day, based on an average daily issuance probability of roughly 1 percent at the CBSA level. Furthermore, responses appear to concentrate in selected states, as is reported in Figure 9, panel B, where the off-days vs. on-days issuance difference is separately estimated by states. In the Appendix, I report the robustness of the results with respect to inclusion of controls and whether consecutive issuance adjustment is applied. Also, I report that the effect is observed mainly in areas with predominantly 1-in-6 day monitoring, while areas with everyday monitoring exhibit an even distribution of Action Days. These findings on coordination in the specific setting of public

²⁶ Prediction of future pollution is based on previous days’ continuous monitoring data. These data are obtained from proxy (i.e. non-regulatory) monitors that are able to provide pollution estimates in real-time. See the Appendix for more detailed discussion of these continuous monitors.

²⁷ CBSAs are urban-centered geographic units representing county groups. Each CBSA has a population size of at least 10,000 and has commuting patterns tied to the urban center. Aggregation of pollution advisories to the CBSA level is motivated by the EPA’s rule which specifies that public broadcast of the Air Quality Index (AQI), which usually serves as the base for Action Day issuances, should be implemented at the CBSA level (U.S. EPA, 2013). I have confirmed that results are not sensitive to using disaggregated reporting agency level data.

behavior manipulation makes it possible to expect that coordination also exists in industrial settings. It appears reasonable to conclude that the observed 1-in-6 day pollution gap may result from a mixture of self-initiated strategic polluting activities and local governments' effort to coordinate polluting activities.

6. Consequences of the Pollution Gap

Thus far, I have shown that the intermittent monitoring policy induces strategic increases in pollution on unmonitored days. In this section, I present evidence that such unintended pollution consequences affect well-being. Specifically, I ask whether health and other economic outcomes are worse on monitor off-days when pollution increases relative to monitor on-days. Put differently, I examine whether outcomes would be better if pollution levels during off-days were consistent with the levels observed on on-days.

As is mentioned in section 3, it is important to note that the full cost of the pollution gap cannot be calculated without knowing the counterfactual level of pollution under daily monitoring. To see this, consider the pollution consequences of “upgrading” from 1-in-6 day monitoring to 1-in-1 day monitoring. It is tempting to think that when polluters face the same incentive to reduce pollution every day that they faced only once per six days before, pollution levels should fall to the previous on-day level. However, I cannot rule out the possibility that such pollution reductions are too costly for the polluters, so that when faced with 1-in-1 day monitoring, the equilibrium average level of pollution is slightly higher than the current on-day level.²⁸

In the following analysis, I document the effect of the pollution gap on standardized test scores and criminal activities, two cognitive and behavioral outcomes that recent literature suggests may be influenced by short-term air pollution fluctuations (e.g., Ebenstein, Lavy, and Roth, 2016; Herrnstadt, Heyes, Muehlegger, and Saberian, 2016). The analysis is a simple extension of the primary examination on pollution, testing whether poorer test performance and higher crime rates coincide with increased pollution levels on off-days. In a companion paper, Zou, Miller, and Molitor (2017) use Medicare administrative data to show that higher pollution on the unmonitored day coincides with a significant increase in elderly mortality rate the following day. Assuming a counterfactual where off-days pollution

²⁸ Notice that pollution should not increase with higher monitoring frequency. Therefore, although it is uncertain by how much will off-day pollution converge to the current on-day level, there is little reason to believe that pollution levels will stay the same or even increase after moving from 1-in-6 day monitoring to 1-in-1 day monitoring.

does not deviate from currently observed on-day levels, Zou, Miller, and Molitor (2017) show that the value of the annual loss of life due to the intermittent monitoring policy is an order of magnitude larger than the resources saved by such a reduced monitoring frequency.

6.1. The Effect of the Pollution Gap on Test Scores

Background and Data. The test scores examination uses the California High School Exit Exams (CAHSEE) school-level test performance data published by the California Department of Education. CAHSEE was designed and first offered in 2001 to volunteer students. Starting 2004, the passage of both CAHSEE math and English tests became an official diploma requirement. Every academic year, multiple test sessions are administered. Each session is comprised of two consecutive test days, with the English test taking place on the first day, and the math test on the second day. The exact administration (such as number of sessions and test months) varies across years, but most sessions took place in July, October, November, December, and February, March, and May in the next calendar year, and so forth. One session, at most, is administered in a given month. For both subjects, test performance is expressed as a scale score that ranges from 275 to 450, with 350 being the usual passing score. All students are required to make their first CAHSEE attempt in grade 10, usually during the February and March sessions (called “Grade 10 census”). Students who miss the census may take makeup sessions held in May. In cases of failure, students are allowed to retake the exams in future sessions (including the census sessions) until both math and English tests were passed. Students are not allowed to retake any exams that they have already passed. As a tradition, most of the tests are scheduled on the first Tuesday or Wednesday of the test month. This scheduling practice creates nice intersection with the EPA’s 1-in-6 day monitoring schedule, which is exploited in the analysis below.

I obtain publicly available test performance data for the universe of tests taken from 2004 to 2013. A unit of observation in the data is a school-test, i.e., a CAHSEE math (English) test taken by all students in a given school on a given date. As a privacy protection measure, the data have a cell size limit of 10 students, so that test scores are masked if they are averaged over the performance of fewer than 10 students. In cases of masked cells, however, I can observe the number of tests taken. The final estimation sample includes test data from about 2,800 schools spanning all 58 counties in California. These test score statistics are aggregated from more than 14 million individual tests taken on 91 different test dates from 2004 to 2013. Table 6 reports test performance statistics. The average math (English) scale score is 367.2

(370.3) within the 275-450 scale. As expected, February and March sessions have much higher average scores due to the fact that tests in other months are mostly taken by students who did not pass CAHSEE on their first attempt in the February or March census. In the analysis below, I control for this compositional difference by including month-of-year fixed effects.

Estimation Framework and Results. As mentioned earlier, because most CAHSEE tests are scheduled on the first Tuesday or Wednesday of the test month, 75 out of 91 tests, i.e., about 4.95 of every 6, coincide with the EPA's monitor 1-in-6 day off-days. However, the scarcity of test dates implies that the interaction between test dates and monitoring dates alone does not ensure strong balance with respect to observable characteristics such as seasonality. In the following, I report two types of specifications. In the first, I use a simple estimation strategy that compares test scores on off-days vs. on-days, including only month-of-year dummies to control for different student composition for tests administered in different months of the year. I then report a much richer specification to ensure balance across various dimensions, controlling for subject fixed effects, school $\times$ month-of-year fixed effects, and time period fixed effects (academic year, day-of-year, weekend). The fixed effects are further interacted with decile distance dummies for the schools' distance to the nearest non-attainment PM monitors.

I begin the analysis with all schools in the data regardless of their location within the state of California. In other words, the analysis starts with a simple comparison of off-days vs. on-days test performance across all schools in California. I then extend the analysis by exploiting geographic variations in the pollution gap, and examine whether test scores responses are stronger for schools that are located closer to non-attainment monitors. In all regressions, I weight observations by the number of tests taken in the school-test cell. Standard errors are two-way clustered at the school and the test levels.

In addition to test scores, I also examine whether the number of tests taken decreases on monitor off-days, as would be the case if pollution causes sickness among students. The econometric specifications are analogous to those used in test scores examination, except that (1) the outcome variable is now log number of tests taken in a school-test cell, and (2) regressions are no longer weighted by the number of tests taken.

Before showing test score results, Table 7 reports the "first stage" estimates of the effect of 1-in-6 day monitoring schedule on air quality. Columns 1 and 2 first show that the 1-in-6 day pollution gap is observed around California high schools. On average, air quality during off-days is roughly 5 percent worse than on on-days. Because comparison of outcome is ultimately done using data on test days, columns 3

and 4 repeat the analysis, restricting the estimation sample to test days. Results show that pollution is about 12 percent worse if the test day coincides with an off-day than if it is an on-day.

Table 8 reports the test performance results. I first focus on the upper panel, where the outcome variable is standardized (i.e., mean 0 and standard deviation 1) test score. Column 1 reports a simple specification where test scores are regressed on an indicator for taking a test on a monitor off-day, conditional on 12 month-of-year dummies that control for differences in the composition of the test takers. This specification shows that taking an exam on an off-day reduces test score significantly, by 5.3 percent of a standard deviation. In column 2, I include the full set of controls as described earlier to ensure balance of various test characteristics across on-days and off-days. Although this specification also yields a negative coefficient estimate, it reduces the effect size to about 2.5 percent of a standard deviation and it is not statistically significant. In columns 3 to 5, I repeat the estimation in column 2, but separately for schools that are close to (< 10 miles) or far away from (10-50 miles and > 50 miles) the nearest non-attainment PM monitor. Results suggest that the effect of off-days is driven by schools close to non-attainment monitors. Column 3 shows that for schools that are within 10 miles of the nearest non-attainment monitor, taking a test on an off-day reduces scores by a statistically significant 6.3 percent of a standard deviation. By contrast, columns 4 and 5 show that no significant responses are detected for schools that are beyond 10 miles from a non-attainment monitor, although the standard errors do not allow me to rule out small effects.

The impact of monitor off-days on test scores is moderate in size. Take the baseline estimate in column 1, for example. On an average off-day, a test score is about 5.3 percent of a standard deviation lower, which is roughly 1.3 points in scale score. The average white-black CAHSEE test score gap from 2004-2013 is 32.15 points. Thus, the effect size of an exam taken on an off-day is about 4 percent of the white-black test score gap. Notably, the effect size is comparable to that from Ebenstein, Lavy, and Roth (2016) who estimate that a standard deviation increase in fine particulate matter leads to a roughly 1.3 percent reduction in matriculation test performance among Israeli students. My pollution and test estimates imply that a standard deviation increase in aerosol pollution corresponds to 2.6 percent reduction in test performance.

In the lower panel of Table 8, I repeat the same estimation, but now test whether fewer tests are being taken on monitor off-days. Note that across columns the sample size is larger than in the test scores regressions because number of tests taken can be observed even for cells where test performance is masked due to privacy protection. Point estimates in columns 1 and 2 suggest that 4.6 percent to 6.8

percent fewer tests are taken on off-days, but the effects are not precisely estimated. Columns 3 through 5 again present estimations separately for schools close to or far away from non-attainment monitors. The analysis provides suggestive evidence of a distance gradient. Column 3 shows that about 11.5 percent fewer tests are taken on an off-day at schools less than 10 miles away from the nearest non-attainment monitor, although the effect is marginally significant. In columns 4 and 5, schools that are farther away show smaller and insignificant responses. Due to the lack of statistical precision, I cannot draw a strong conclusion that fewer students attend tests administered on monitor off-days.

6.2. The Effect of Pollution Gap on Crime

Background and Data. I use crime data from the Federal Bureau of Investigation's National Incident-Based Crime Reporting System (NIBRS) from 2001 to 2013. These data contain detailed crime incident-level information, such as the date, the location of the reporting jurisdiction, and the offense code, reported by jurisdictions that participate in the NIBRS program. Reporting jurisdictions are usually city (or county) law enforcement agencies. The number of participating jurisdictions in the NIBRS has grown over time. By 2013, NIBRS covered a population of about 92 million people in 33 states, and accounted for more than 28 percent of all crime reported to the FBI Uniform Crime Reporting Program.

Despite the growing coverage, crime prevalence in NIBRS-participating jurisdictions is not considered to be representative of overall crime rates. For example, jurisdictions with fewer people are known to be disproportionately more likely to report data (James and Council, 2008). Differences in levels of crime are not necessarily a threat to the identification strategy which uses day-to-day variation in pollution monitoring status within the same area. To alleviate concerns about compositional changes in the NIBRS-covered population over time, my analysis is restricted to jurisdictions that have participated in NIBRS for at least 10 years during the study period.²⁹ However, I recognize that sensitivity to air pollution may differ among people living in reporting and non-reporting areas. This is an important caveat regarding external validity throughout the interpretation of results obtained in this section.

I begin by first constructing daily crime rates at the county level using information on jurisdictions' county location and the population covered. The crime rate is defined as the reported number of crime

²⁹ In unreported results, I confirm that the findings in this section are similar if I relax the restriction on panel balance, or if I use a strictly balanced panel, i.e. counties that consistently report to NIBRS for the entire 2001-2013 period.

incidents in the county divided by the population covered by NIBRS within the county.³⁰ These data are then merged with counties' PM monitoring frequencies. From 2001 to 2013, 356 counties follow the 1-in-6 day sampling schedule (defined as counties where all PM monitors follow the 1-in-6 day schedule) and crime data are available in 47 counties. These counties span 19 different states, and include a total population of 1.33 million people, representing 67 percent of the total population in those counties.

My analysis focuses on three broad groups of criminal activities. Following the FBI's categorization, I create crime rate variables for violent crime (aggravated assault, robbery, forcible rape, murder, and nonnegligent manslaughter), property crime (burglary, larceny-theft, motor vehicle theft, and arson), and all other crime. Table 9 presents summary statistics for counties in the estimation sample, i.e. counties that follow the 1-in-6 day schedule and with crime data available. These statistics are compared to statistics of the entire NIBRS population.

Estimation Framework and Results. The identification of the causal effect of the monitoring schedule on crime is once again a straightforward comparison of crime rates on monitor off-days vs. on-days. As before, I report results from two polar specifications: one with no controls, and the other with a full array of fixed effects controls (county, year, month-of-year, day-of-week, and day-of-month). Regressions are weighted by the population covered by NIBRS in the county. Standard errors are clustered at the county level.

Table 10, column 1 begins by repeating the pollution analysis. The estimation conceptually corresponds to column 4 of Table 2, whereas now pollution is measured at the county level, i.e., the average of all 10km × 10km pixels within the county border. For example, column 1, panel A shows that the off-days vs. on-days pollution gap is estimated to be 1.3 percent, which is smaller than the effect size observed at the monitoring sites (a 1.8 percent pollution gap). This is likely due to the fact that gaming is targeting at the monitoring sites, and therefore the average effect observed at the county level is lower. In column 2, the pollution regression is repeated, again using the estimation sample, which yields a pollution gap of 2.7 percent without controls and 1.9 percent with controls. The 95 percent confidence intervals of these estimates overlap with the mean coefficients in column 1.

³⁰ Every year, about 10 percent of incidents of crime occur in jurisdictions that cross county borders. In these cases, NIBRS provides estimates on the population that is covered by the jurisdiction in each of the counties the jurisdiction spans. I use this information to assign a given incident of crime to each of the counties, with the assignment probability proportional to the county's population share within the jurisdiction.

Columns 3 to 5 document crime effects of the monitoring schedule. I first focus on panel A, which presents raw comparison of off-days vs. on-days means. Column 3 shows that raw comparison across off-days vs. on-days means show that violent crime is about 0.257 per million (NIBRS population) higher on off-days. Based on the daily mean of 15.88 per million, the effect represents a 1.6 percent increase. Whereas previous literature documents little evidence on the effect of pollution on property crime, column 4 reports that property crime, including burglary, larceny-theft, motor vehicle theft, and arson, also increases by about 1 per million on an off-day. However, the relative size of this effect is smaller than the violent crime effect -- about 0.91 percent out of the daily mean of 110.31 per million. Column 5 continues the analysis with other categories of crime, in which I find no evidence of a precise increase. Panel B repeats the same analysis controlling for geographic and time fixed effects, which yields very similar results.

7. Conclusion

Fiscal constraints often motivate environmental regulators to monitor polluting behavior on an intermittent basis. A largely overlooked issue with intermittent monitoring is its vulnerability to polluters' strategic responses, as is highlighted by recent evidence on the vehicle emission scandals. This paper reinforces these recent findings and extends the literature to a broader setting of ambient air quality regulation. I have presented evidence that a widely used once-every-six-day monitoring schedule for outdoor particle pollution causes significant deterioration in air quality on unmonitored days as compared to the levels observable to the air regulator on monitored days.

To shed light on the mechanisms by which strategic responses may arise, I show that the intensity of responses differs substantially across geographic regions, depending strongly on the characteristics of local areas' economies and industries. Overall, my results are consistent with the view that strategic responses are more likely to arise where the *incentives* to avoid monitoring are stronger, such as areas subject to non-compliance punishments, and where the *scope* for avoidance is larger, such as areas with high-emission industries that often do not operate on full capacity. In addition, my results reveal the possibility that strategic responses may arise through coordination by the local governments who, by design under the federal regulation, share non-compliance costs. As an illustration of this mechanism, I show that some local governments issue air quality advisories strategically in response to the monitoring

schedule, an action that is likely intended to manipulate public behavior such as transportation decisions, and, ultimately, to reduce pollution levels recorded on monitored days.

Not only do strategic responses compromise the effectiveness of regulation enforcement, the unintended pollution consequences may also cause damage to well-being. This paper has documented that the strategic increase in pollution on monitor off-days has led to lower performance on academic tests and to higher crime rates. In a companion paper, Zou, Miller, and Molitor (2017) present evidence that the higher levels of pollution experienced on unmonitored days also leads to increased mortality rates among the elderly on these days. Calculating the damage of these consequences is difficult without precise information on the level of pollution under a counterfactual continuous monitoring policy. However, if the pollution levels on unmonitored days were consistent with the levels observed on monitored days, then these outcomes could have been significantly improved.

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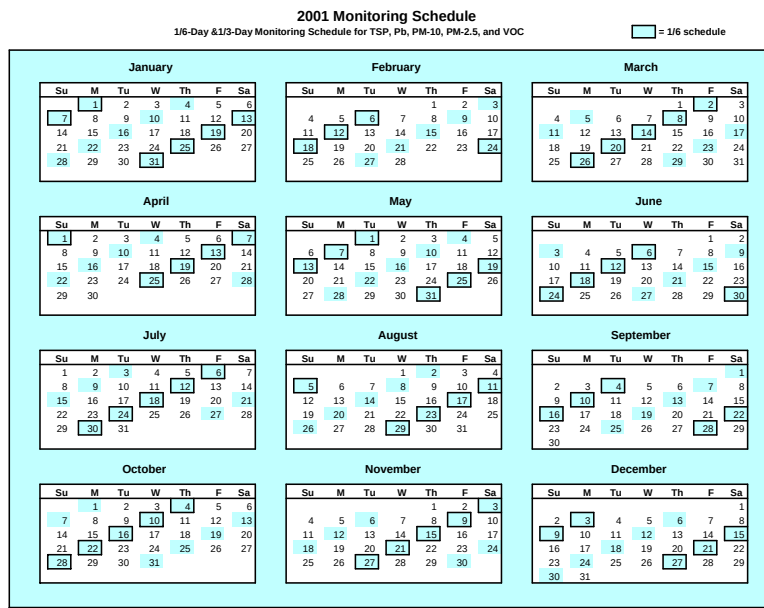
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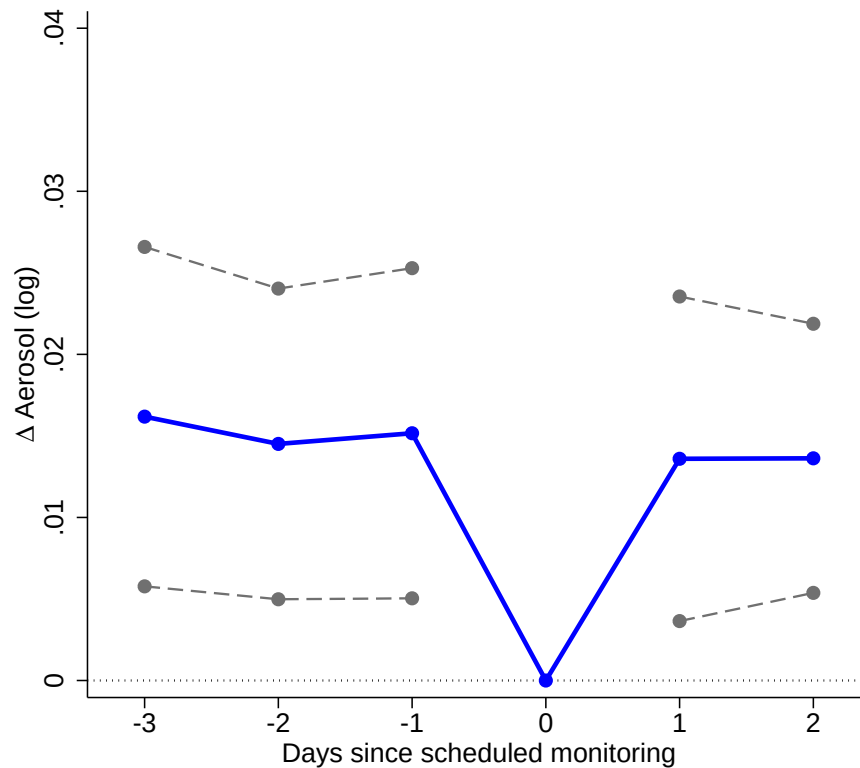
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Figure 1. EPA's Ambient Pollution Monitoring Schedule, 2001



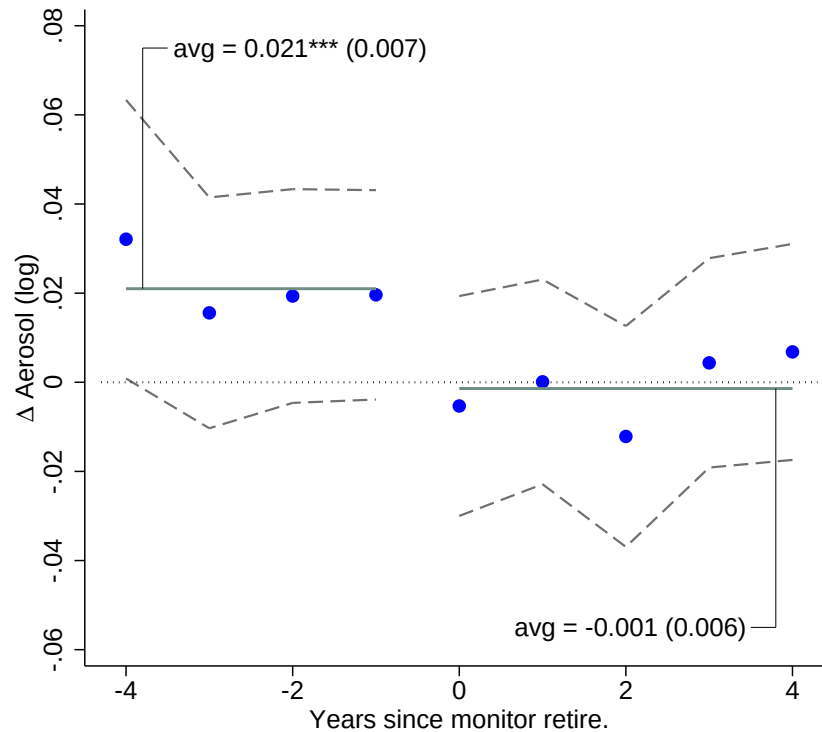
Notes: Figure shows the EPA's 2001 monitoring schedule calendar. Full archives of all calendars can be found here: <<https://www3.epa.gov/ttn/amtic/calendar.html>>.

Figure 2. Event Study: Off-days vs. On-days Pollution Gap



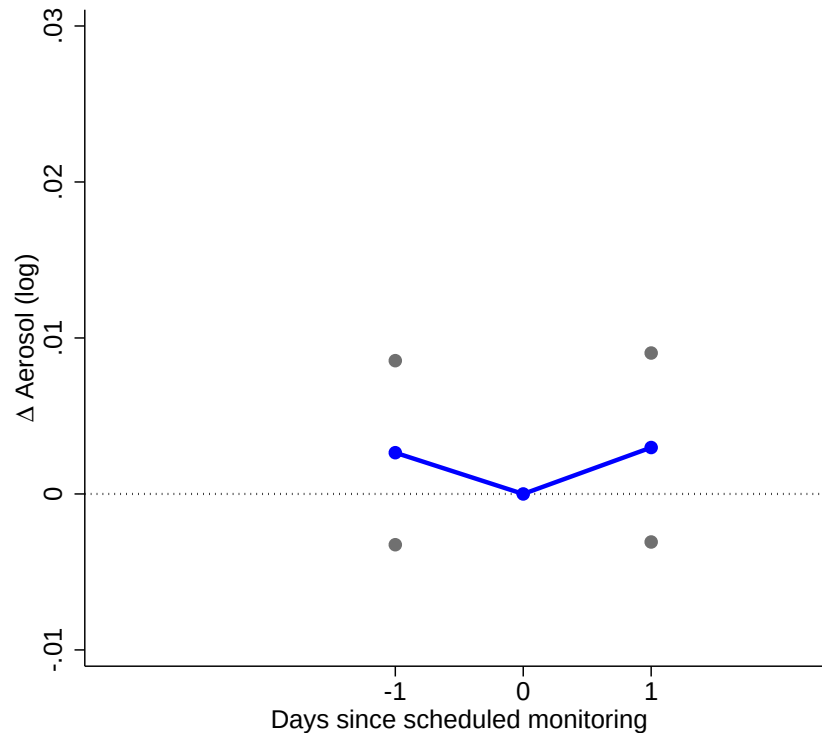
Notes: Figure plots the path of pollution concentration by days of 1-in-6 day monitoring cycle. Sample includes all sites that contain at least one 1-in-6 day PM monitor. Pollution is measured by satellite-based aerosol concentration within the 10km×10km area that contains the monitoring site. Day 0 corresponds to the scheduled sampling day which is normalized to 0. The regression is not conditional any covariates. Dashed lines represent 95% confidence intervals constructed using standard errors clustered at the county level.

Figure 3. Placebo Test: Pollution Gap by Years Relative to Monitoring Site Retirement



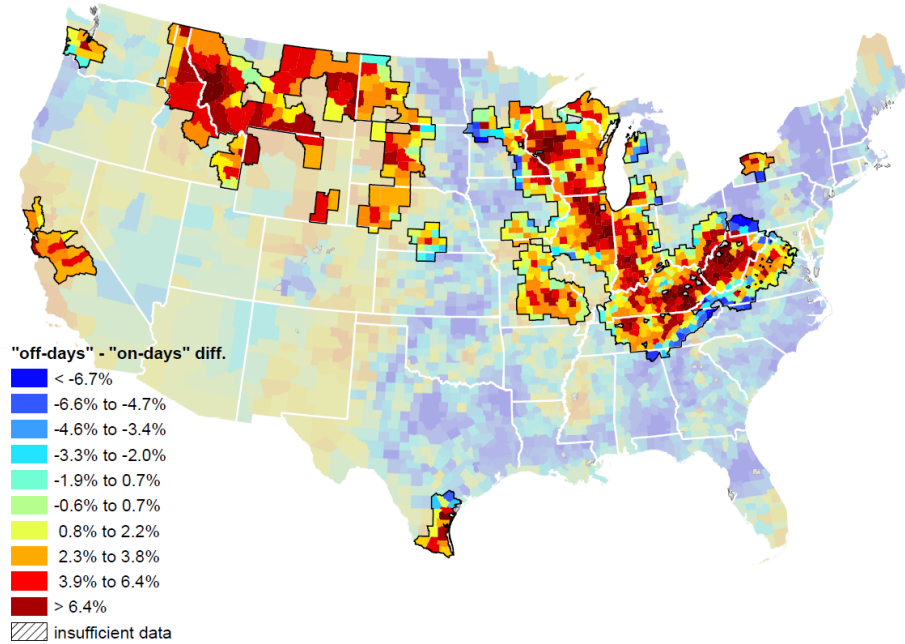
Notes: Figure plots off-days vs. on-days pollution gap as a function of years relative to site retirement. Average estimates show off-days effect separately estimated before and after monitoring site retirement. Sample includes retirement of 490 sites from 2001 to 2013. The regression is not conditional on any covariates. Dashed lines represent 95% confidence intervals constructed using standard errors clustered at the county level.

Figure 4. Event Study: Pollution Gap Near 1-in-3 Day Monitoring Sites



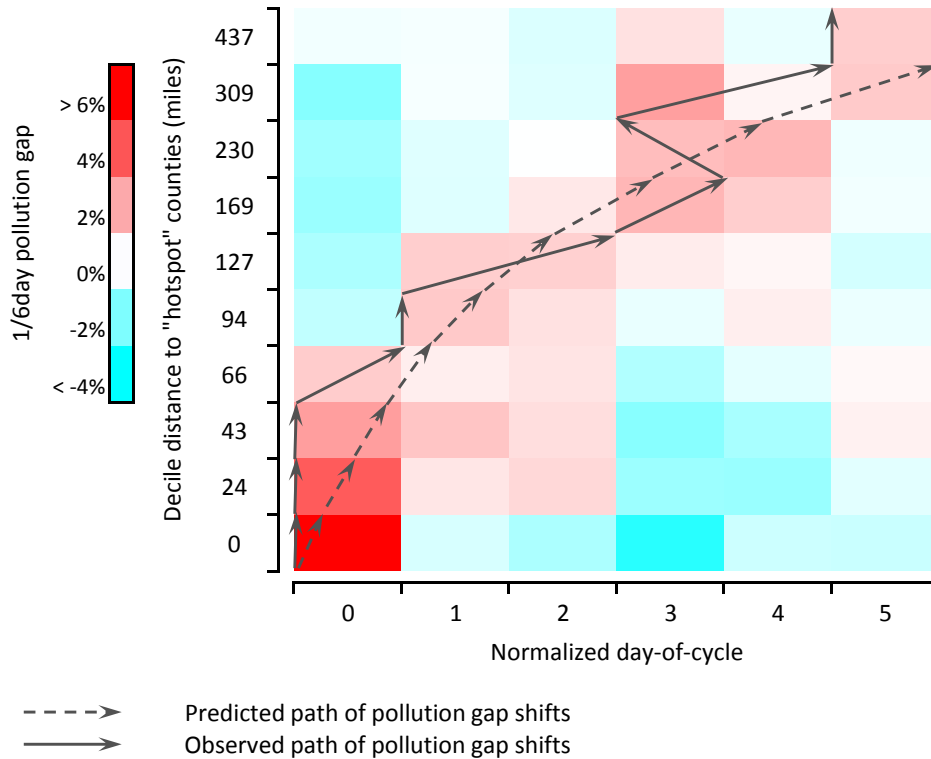
Notes: Figure plots the path of pollution concentration by days of 1-in-3 day monitoring cycle. Sample includes all sites that contain at least one 1-in-3 day PM monitor. Pollution is measured by satellite-based aerosol concentration within the 10km $\times$ 10km area that contains the monitoring site. Day 0 corresponds to the scheduled sampling day which is normalized to 0. The regression is not conditional on any covariates. Dashed lines represent 95% confidence intervals constructed using standard errors clustered at the county level.

Figure 5. County Level 1-in-6 Day Pollution Gap



Notes: Map plots county-level 1-in-6 day pollution gap estimates. Red (blue) areas correspond to higher (lower) aerosol levels during off-days. For all counties, off-days and on-days are defined using EPA's monitoring schedule. The "Insufficient data" category corresponds to counties with fewer than 6,000 available aerosol observations. Map also highlights "hot-spot" counties, i.e. those in the top decile of pollution gap magnitude, along with counties within 50 miles to the hot-spot counties.

Figure 6. Pollution Gap Shifts



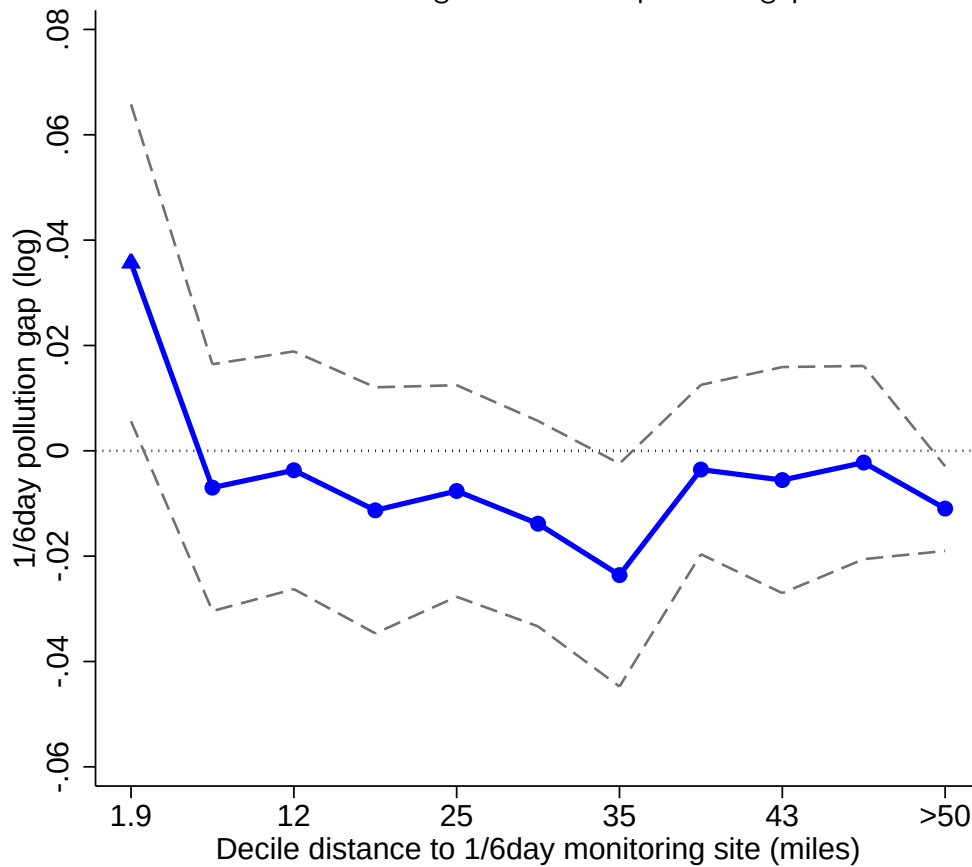
Notes: Each row represents a group of counties by their decile distance to the hot-spot counties. Each column represents a day in a 1-in-6 day monitoring cycle, with 0 being the on-day according to the EPA's monitoring schedule. Each cell shows the (log) pollution difference between that day and the other five days on the same row (so, for column day-0, this is just the off-day vs. on-day pollution gap). The underlying estimation data contains 128,065,517 observations at the 10km×10km satellite pixel × daily level. Regression models are estimated separately for each row. Dashed arrows show shifts of the timing when minimum pollution level is *expected* to be observed within a six-day monitoring cycle when directivity-weighted wind speed is 3.03 miles/hour. Solid arrows show the *observed* timing of minimum pollution level.

Figure 7. Industry Correlates of Pollution Gap Hotspots

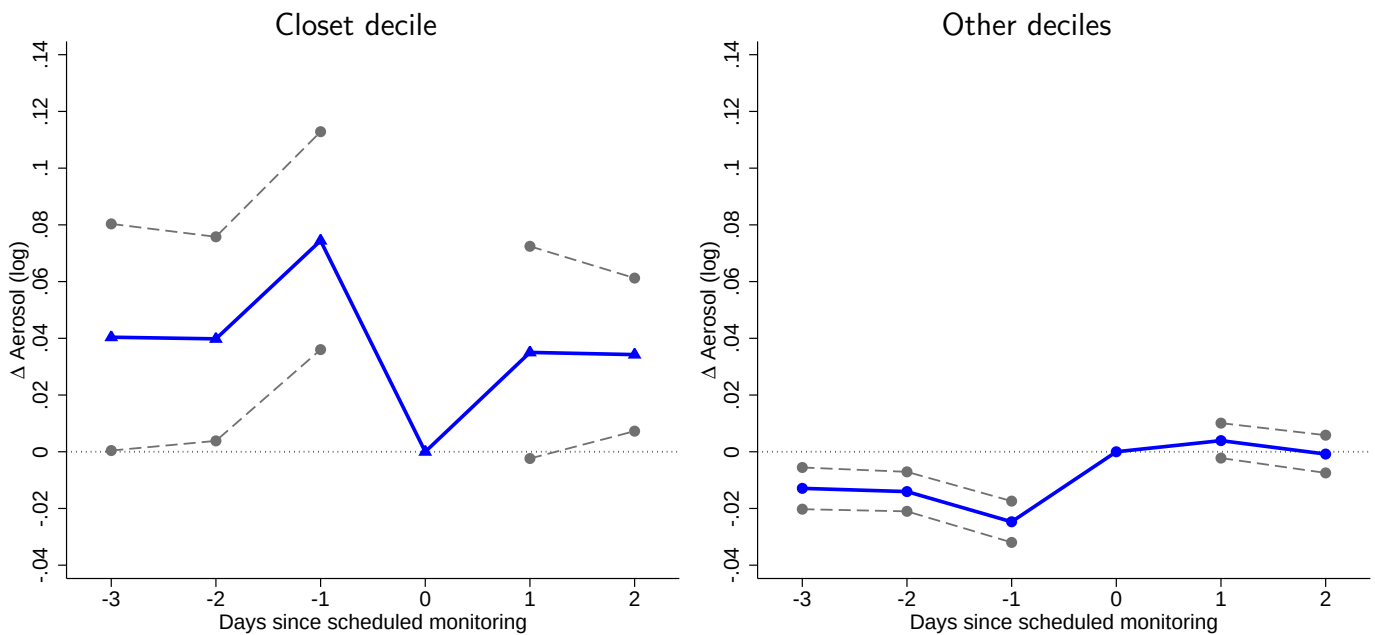


Notes: Each column represents t -statistics from a single regression, with model specifications laid out in the header. 2-digit NAICS industry blocks are ordered by its share of national total PM_{10} emissions according to EPA's 2011 National Emissions Inventory. In the interest of space, graph only reports results for polluting sectors, i.e. those contribute at least 1% of total PM_{10} . Redness (blueness) indicates that the county being an industrial hot spot positively (negatively) predicts the likelihood of the county being a 1-in-6 day pollution gap hot spot.

Figure 8. Pollution Gap Around Wood Product Manufacturing Plants
Panel A. Distance gradient of the pollution gap

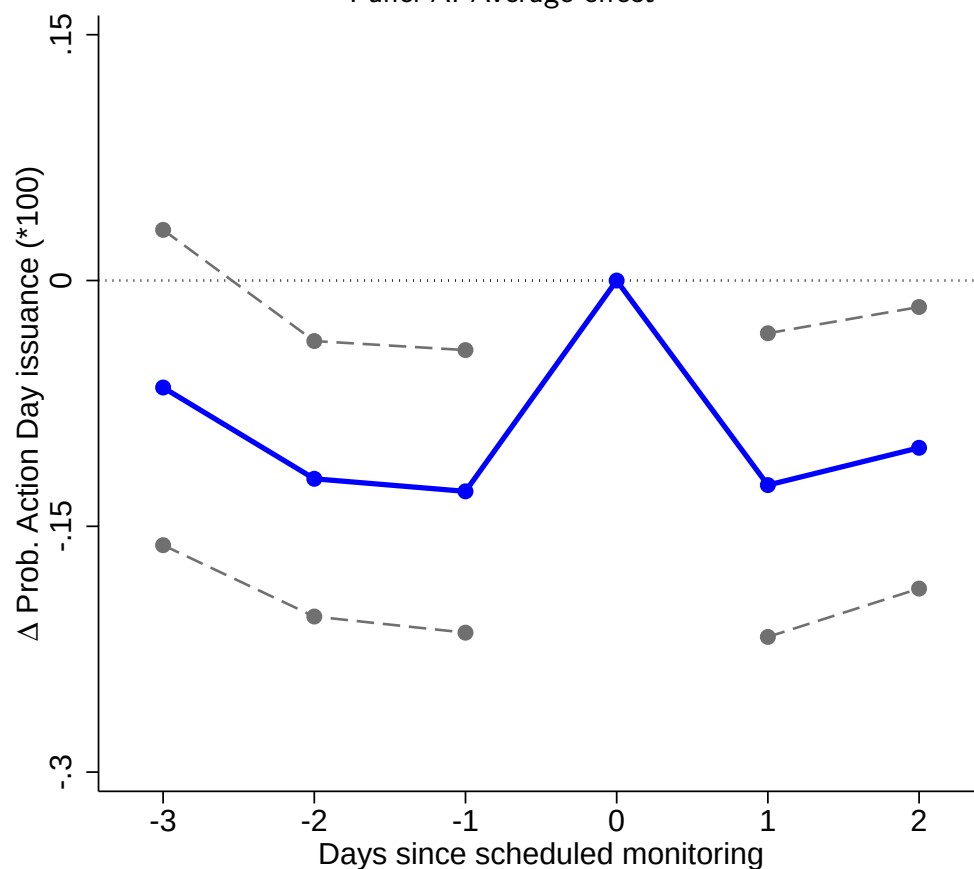


Panel B. Pollution gap for plants near vs. far-away from 1-in-6 day monitors

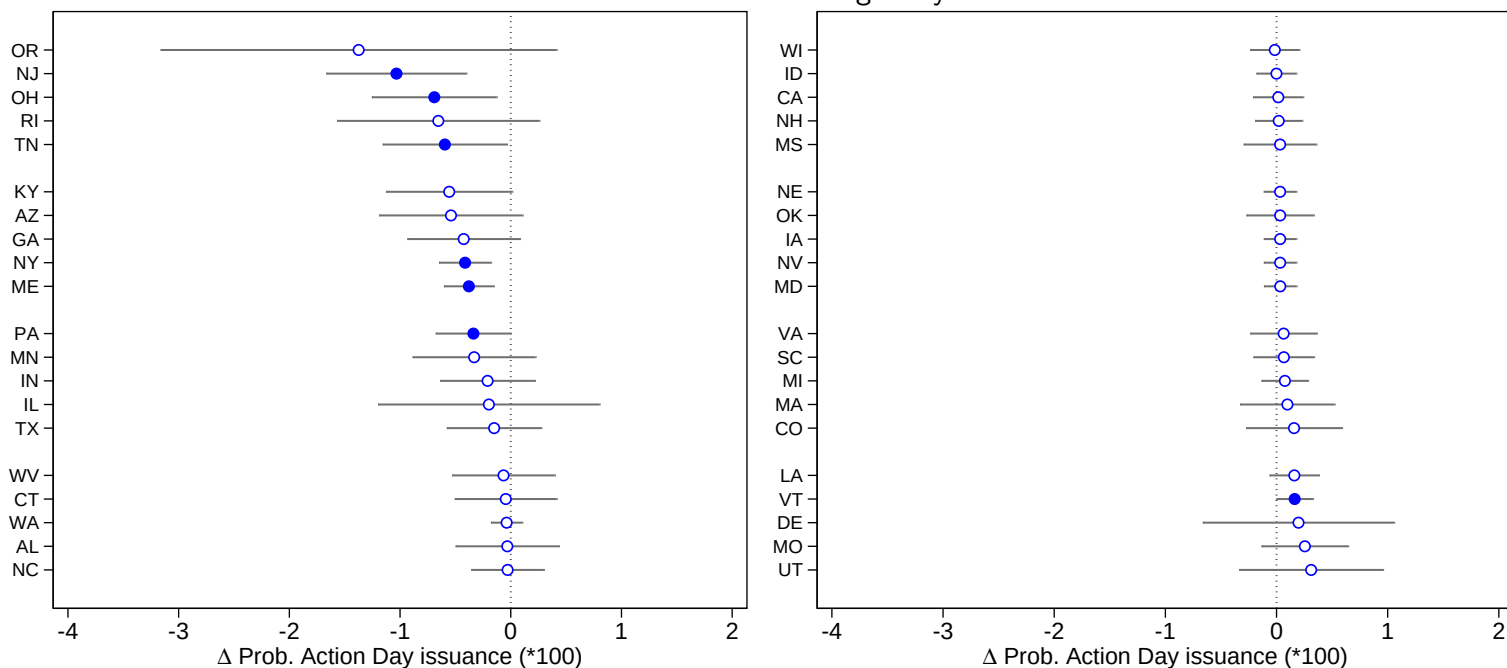


Notes: Sample includes all wood product manufacturing plants (NAICS = 321) observed in the EPA's annual Toxic Release Inventory from 2001-2013. The underlying estimation data contains 610,459 observations at the 10km×10km satellite pixel × daily level. Panel A plots 1-in-6 day pollution gap around plants that are within 50 miles to the nearest 1-in-6 day PM monitoring site, grouped by decile distance. The rightmost bin groups all facilities that are more than 50 miles apart from the nearest site. Panel B shows 1-in-6 day pollution patterns for close and far-away plants. Dashed lines represent 95% confidence intervals constructed using standard errors clustered at the county level.

Figure 9. Local Governments' Strategic "Pollution Action Day" Declaration
Panel A. Average effect



Panel B. State heterogeneity



Notes: Outcome variable is core based statistical area (CBSA) $\times$ daily dummy for weather any action day is issued. Sample spans 2004-2013 and includes 14,945 issuances across 171 CBSAs. In cases of issuances that span a consecutive number of days, only the first day of issuance is counted. See text for more details. Dashed lines represent 95% confidence intervals constructed using standard errors clustered at the CBSA level. Panel A reports average off-days vs. on-days probability across all CBSAs that ever issued any action days during the sample period. Panel B shows off-days vs. on-days issuance probability differential estimated separately for each state.

Table 1. Summary Statistics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
	aerosol stats ($\times 100$)			Total observations	monitor stats			site stats		<i>N</i>	county stats		
Year	p25	mean	p75		<i>N</i>	<i>N</i> 1/6day	<i>N</i> 1/6day > NAAQS	<i>N</i>	<i>N</i> 1/6day		pop (million)	<i>N</i> 1/6day	pop 1/6day (million)
2001	3.0	11.4	15.6	9,337,915	1,816	876	52	1,343	776	697	193.2	429	142.6
2002	3.3	14.1	18.1	9,301,308	1,933	889	57	1,413	774	725	206.3	428	143.3
2003	3.2	12.6	17.0	9,513,247	1,775	831	63	1,322	725	676	200.3	399	142.4
2004	2.6	10.7	14.7	9,322,004	1,865	850	25	1,377	737	694	204.9	403	149.6
2005	2.8	12.2	16.1	10,160,251	1,792	785	60	1,311	679	653	199.7	362	141.2
2006	3.2	12.4	16.4	10,174,374	1,817	828	86	1,296	690	657	202.8	368	141.2
2007	3.7	14.5	18.7	10,295,391	1,761	740	126	1,262	600	652	208.2	322	138.8
2008	3.7	12.3	16.7	10,228,175	1,628	663	73	1,158	538	606	204.6	287	133.9
2009	3.8	11.4	15.7	9,469,757	1,728	681	39	1,199	537	633	210.1	297	132.2
2010	3.3	10.5	14.2	10,078,748	1,702	659	38	1,159	522	619	209.3	285	132.0
2011	4.5	13.4	17.2	10,298,153	1,585	579	39	1,077	458	578	200.0	262	124.8
2012	4.0	13.3	17.8	10,806,884	1,629	540	20	1,088	426	582	195.4	253	114.5
2013	3.0	11.6	15.6	9,079,310	1,699	522	22	1,109	406	596	203.4	229	112.9

Notes: Each row represents statistics for a calendar year. Column 1, 2, and 3 shows 25th-percentile, mean, and 75th-percentile values of grid-daily level aerosol concentration. Column 4 shows total number of grid-daily observations. Monitor sample includes all monitors that collected enough samples during the year to be considered eligible for NAAQS comparison. Column 5 reports total number of particulate matter (PM) monitors, including all PM_{2.5} and PM₁₀ monitors that are eligible for NAAQS comparison (see the text for more details). Column 6 counts number of 1-in-6 day monitors, defined by monitors that are required to sample either 60 or 61 days of PM data for each calendar year. Column 7 counts number of 1-in-6 day monitors that exceeded any PM standard in that year (but not necessarily violating the NAAQS, as violation is based on 3-year average values). Column 8 aggregates monitor counts from column 1 to the monitoring site level, acknowledging the fact that there might be multiple PM monitors within the same monitoring sites. Column 9 counts number of sites that contain at least one PM monitor that follows the 1-in-6 day schedule. Column 10 and 12 aggregate site counts in column 8 and 9 to the county level, respectively. Column 11 and 13 report corresponding population that lived in the monitored counties.

Table 2. Off-days vs. On-days Pollution Gap: 1-in-6 Day Monitoring Sites				
Dep. var. = Aerosol concentration (log)				
	(1) Sample: sites w. any 1/6d monitor	(2) Sample: sites w. any 1/6d monitor	(3) Sample: sites w. only 1/6d monitor	(4) Sample: counties w. only 1/6d monitor
1(off-days)	0.016*** (0.004)	0.016*** (0.004)	0.018*** (0.004)	0.018*** (0.006)
Ctrls		✓	✓	✓
N	685,060	685,060	427,846	176,225
N (site)	1,193	1,193	899	489

Notes: Each column reports a separate regression. Column name indicates the sample used. Column 1 & 2 use all sites that have at least one 1-in-6 day PM monitor. Column 3 includes sites that have standalone 1-in-6 day monitor. Column 4 includes sites in counties with only 1-in-6 day monitors. Controls include fixed effects dummies (site, year, month-of-year, and day-of-week) and weather controls. Standard errors are clustered at the county level. *: $p < 0.10$; **: $p < 0.05$; ***: $p < 0.01$.

Table 3. Off-days vs. On-days Pollution Gap: "Placebo" Sites			
Dep. var. = Aerosol concentration (log)			
	(1) Sample: retired 1/6d sites	(2) Sample: 1/1d sites	(3) Sample: Non-PM 1/6d sites (HAPs)
1(off-days)	-0.0020 (0.0046)	0.0023 (0.0080)	0.0023 (0.0044)
Power _(1.5% effect, 5% sig.)	0.940	0.803	0.910
N	372,989	231,532	370,020
N (site)	490	556	792

Notes: Each column reports a separate regression. Column name indicates the sample used. Column 1 includes areas that had 1-in-6 day PM monitoring sites which retired. Column 2 includes 1-in-1 day sites. Column 3 includes 1-in-6 day HAPs sites. Controls include fixed effects dummies (site, year, month-of-year, and day-of-week) and weather controls. Power calculation estimates power of tests detecting a 1.5% mean difference between off-days vs. on-days at a 5% significance level. Standard errors are clustered at the county level. *: $p < 0.10$; **: $p < 0.05$; ***: $p < 0.01$.

Table 4. Off-days vs. On-days Pollution Gap: 1-in-3 Day Monitoring Sites				
Dep. var. = Aerosol concentration (log)				
	(1) Sample: sites w. any 1/3d monitor	(2) Sample: sites w. any 1/3d monitor	(3) Sample: sites w. only 1/3d monitor	(4) Sample: counties w. only 1/3d monitor
1(off-days)	0.0028 (0.0026)	0.0029 (0.0020)	0.0024 (0.0025)	0.0054* (0.0030)
Ctrls		✓	✓	✓
<i>N</i>	598,859	598,859	386,854	244,071
<i>N</i> (site)	1,064	1,064	849	562

Notes: Each column reports a separate regression. Column name indicates the sample used. Column 1 & 2 use all sites that have at least one 1-in-3 day PM monitor. Column 3 includes sites that have standalone 1-in- day monitor. Column 4 includes sites in counties with only 1-in-3 day monitors. Controls include fixed effects dummies (site, year, month-of-year, and day-of-week) and weather controls. Standard errors are clustered at the county level. *: $p < 0.10$; **: $p < 0.05$; ***: $p < 0.01$.

Table 5. Correlates of Pollution Gap Hotspots: Compliance Status and Emission Concentration

Dep. var. = 1(hot-spot counties). Mean = 0.10					
	(1)	(2)	(3)	(4)	(5)
1(Non-attainment)	0.052*** (0.018)			0.042** (0.019)	0.046** (0.019)
1(1-in-6 day monitoring)		0.039*** (0.014)		0.029* (0.015)	0.001 (0.017)
1(Emission Herfindahl ≥ 0.9)			0.004 (0.011)	0.007 (0.011)	-0.007 (0.012)
1(Emission Herfindahl ≥ 0.9) × 1(1-in-6 day monitoring)					0.109*** (0.033)
<i>N</i>	3,199	3,199	3,199	3,199	3,199

Notes: Each column reports a separate regression. A unit of observation is a county. Dependent variable is a dummy which equals one if a county is among the “hot-spot” counties, i.e. those in the top decile of pollution gap magnitude, and therefore has a mean value of roughly 0.10. “1(Non-attainment)” indicates counties ever designated PM₁₀ or PM_{2.5} non-attainment status. “1(Has 1-in-6 day sites)” indicates counties where any 1-in-6 day PM monitors existed. “1(Emission Herfindahl ≥ 0.9)” indicates counties where the 2001-2013 annual average emission Herfindahl Index is larger than 0.9, based on the Toxic Release Inventory. Standard errors are unclustered. *: $p < 0.10$; **: $p < 0.05$; ***: $p < 0.01$.

Table 6. Summary Statistics: California High School Exit Exam (CAHSEE) Scores

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	English				Math			
	Overall	Feb-Mar	May	Other	Overall	Feb-Mar	May	Other
Scale score (275-450)	367.2 [24.2]	375.2 [21.2]	338.2 [14.1]	341.8 [10.8]	370.3 [24.7]	378.3 [21.9]	342.2 [12.4]	344.1 [10.3]
Number of tests	52	15	10	27	59	19	7	33
Number of tests taken	8,370,191	6,411,603	403,451	1,555,137	8,301,056	6,383,758	388,841	1,528,457

Notes: Statistics are computed from school-subject-daily level data. Column 1 to 4 (5 to 8) present statistics for English (math) tests. Column 1 and 5 report overall statistics. The remaining columns report statistics by month of test administration. Standard deviations are reported in brackets.

Table 7. 1-in-6 day Pollution Gap Around High Schools

Dep. var. = Aerosol concentration (log)				
	(1)	(2)	(3)	(4)
	Sample: All schools All days		Sample: All schools Exam days	
1(off-days)	0.049*** (0.003)	0.053*** (0.002)	0.128*** (0.036)	0.117*** (0.034)
Ctrls.		✓		✓
N	825,800	825,800	41,630	41,617

Notes: Each cell corresponds to a separate regression. Regressions in column 1 and 2 include all schools on all days from 2001-2013, comparing aerosol concentration on off-days vs. on-days. Column 3 and 4, comparing aerosol concentration during off-days CAHSEE tests and on-days CAHSEE tests. Aerosol concentration at the school is measured at the 10km×10km grid cell area that contains the school. Controls include grid cell, year, month-of-year, and day-of-week FEs. *: $p < 0.10$; **: $p < 0.05$; ***: $p < 0.01$.

Table 8. 1-in-6 day Gap: Standardized Test Performance (CAHSEE, California)

	(1)	(2)	(3)	(4)	(5)
	Sample:		Sample:		
	All schools		Schools close to natt. sites		
			0-10 miles	10-50 miles	>50 miles
Panel A. Dep. var. = Scale score (std.)					
1(off-days)	-0.053** (0.024)	-0.025 (0.026)	-0.063** (0.032)	-0.009 (0.030)	0.011 (0.024)
<i>N</i>	122,540	116,922	34,537	45,948	36,429
<i>N</i> (dates)	91	91	91	91	91
Panel B. Dep. var. = Number of tests taken (log)					
1(off-days)	-0.046 (0.052)	-0.068 (0.046)	-0.115* (0.060)	-0.084 (0.052)	-0.011 (0.042)
<i>N</i>	206,519	189,555	49,296	75,650	64,602
<i>N</i> (dates)	91	91	91	91	91
Month-of-year ctrls.	✓				
Full ctrls.		✓	✓	✓	✓

Notes: Each cell corresponds to a separate regression. Regressions in column 1 and 2 include all schools. Column 3, 4 and 5 break down the sample by the distance between the school to the closest non-attainment monitor. In the upper panel, outcome variables are standardized scale scores and logged number of test takers. In the lower panel, outcome variable is logged aerosol level at the school level. Standard errors are two-way clustered at the school and the exam date level. *: $p < 0.10$; **: $p < 0.05$; ***: $p < 0.01$.

Table 9. Summary Statistics: Daily Crime Rates

	(1)	(2)
	Estimation sample	NIBRS population
Violent crime (per million)	15.88 [14.99]	11.93 [23.54]
Property crime (per million)	110.31 [55.26]	97.92 [59.87]
Other crime (per million)	118.33 [69.72]	92.36 [116.11]
Number of counties	47	404

Notes: Column 1 presents statistics for counties included in the estimation, i.e. counties with 1-in-6 day sampling and reported crime data for at least 10 years to NIBRS over the 2001-2013 period. Column 2 presents statistics for all counties that ever reported to NIBRS during the same period. Standard deviations are reported in brackets.

Table 10. 1-in-6 day Gap: Crime

	(1) Sample:	(2)	(3)	(4) Sample:	(5)
	Counties w. 1/6d sampling	NIBRS counties w. 1/6d sampling			
Dep. var.	Aerosol (log)	Aerosol (log)	Violent crime (per million)	Property crime (per million)	Other crime (per million)
Panel A. No ctrls.					
1(off-days)	0.013*** (0.004)	0.027** (0.013)	0.257** (0.104)	1.004** (0.426)	0.284 (0.417)
Panel B. Full ctrls.					
1(off-days)	0.009** (0.004)	0.019 (0.011)	0.255** (0.110)	0.960** (0.416)	0.255 (0.417)
Mean dep. var.			15.88	110.31	118.33
N	224,847	25,981	68,666	68,666	68,666
N (county)	356	47	47	47	47

Notes: Each cell corresponds to a separate regression. Estimation samples restrict to counties with no high frequency PM monitors. Column 1 replicates the main pollution regression at the county level. Column 2 is the pollution regression restricting to counties included in the crime regressions. Column 3 to 5 present crime regression results. Panel A reports estimation with no covariates. Panel B reports estimation with full set of controls including county fixed effects, year fixed effects, month-of-year fixed effects, day-of-week fixed effects, and day-of-month fixed effects. Standard errors are clustered at the county level. *: $p < 0.10$; **: $p < 0.05$; ***: $p < 0.01$.